REVIEW

INNOVATION AND EMERGING TECHNOLOGIES

Large language models in healthcare and biomedical informatics: A comprehensive review

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Large language models (LLMs) are rapidly advancing natural language processing and driving increasing interest across biomedical research, clinical care, and healthcare operations. This systematic review synthesizes current empirical evidence on LLM applications in healthcare and biomedical informatics, focusing on their capabilities, evaluation practices, and reported outcomes. Existing studies highlight promising uses in research, where LLMs assist literature-informed genomic interpretation, functional annotation, and biological hypothesis generation, as well as support tasks related to protein and multiomics analysis. In clinical contexts, LLMs are primarily evaluated for natural language-driven tasks, including electronic health record summarization, clinical documentation support, medical question answering, and information extraction, rather than autonomous diagnostic or therapeutic decision-making. Early investigations also describe potential value in healthcare operations, patient communication, clinician education, and drug discovery workflows largely through knowledge retrieval, text generation, and semantic search. However, current evidence remains preliminary: most studies are retrospective, benchmark-based, simulation-driven, or limited to controlled research settings. Key challenges include hallucination and unreliable reasoning, bias and inequitable performance across populations, data privacy and security constraints, reproducibility limitations, and lack of prospective clinical validation and regulatory guidance. Emerging strategies, such as domain-specific pretraining, retrieval augmentation, multi-modal architectures, and privacy-preserving learning, aim to improve reliability, safety, and real-world applicability. This review concludes by outlining methodological, infrastructural, and governance requirements needed to responsibly integrate LLMs into biomedical workflows, emphasizing that clinical deployment remains exploratory and must be rigorously evaluated.

***Keywords*: Bioinformatics; Healthcare; Large Language Models; Natural Language Processing.**

INTRODUCTION

Natural language processing (NLP) has long been essential in healthcare and biomedical informatics, where large volumes of unstructured clinical notes, radiology reports, discharge summaries, and biomedical literature contain critical information for patient care, research, and operations[1]. Early rule-based and statistical NLP systems enabled tasks such as named entity recognition (NER), concept extraction, and document classification, supporting applications in electronic health records (EHR) management, cohort identification, and literature curation[2].

Traditional approaches, however, struggled with the complexity of medical language, including domain-specific terminology, abbreviation ambiguity, and contextual interpretation of diagnoses and treatments. General-purpose NLP models also failed to capture the nuanced semantics and hierarchical structures of biomedical text, limiting accuracy and generalizability[3]. Deep learning (DL) approaches, particularly recurrent neural networks (RNNs) and long short-term memory (LSTM) models, improved sequential modeling but remained limited by difficulty capturing long-range dependencies and inefficiency in parallelization. Transformer-based models addressed these issues by replacing recurrence with self-attention, enabling efficient training and rich contextual representations[4].

Bidirectional encoder representations from transformers (BERT) advanced the field by introducing masked language modeling and transfer learning, allowing a single pretrained model to be fine-tuned across diverse tasks[5]. Domain-specific models such as BioBERT, trained on large biomedical corpora, achieved state-of-the-art (SOTA) results in

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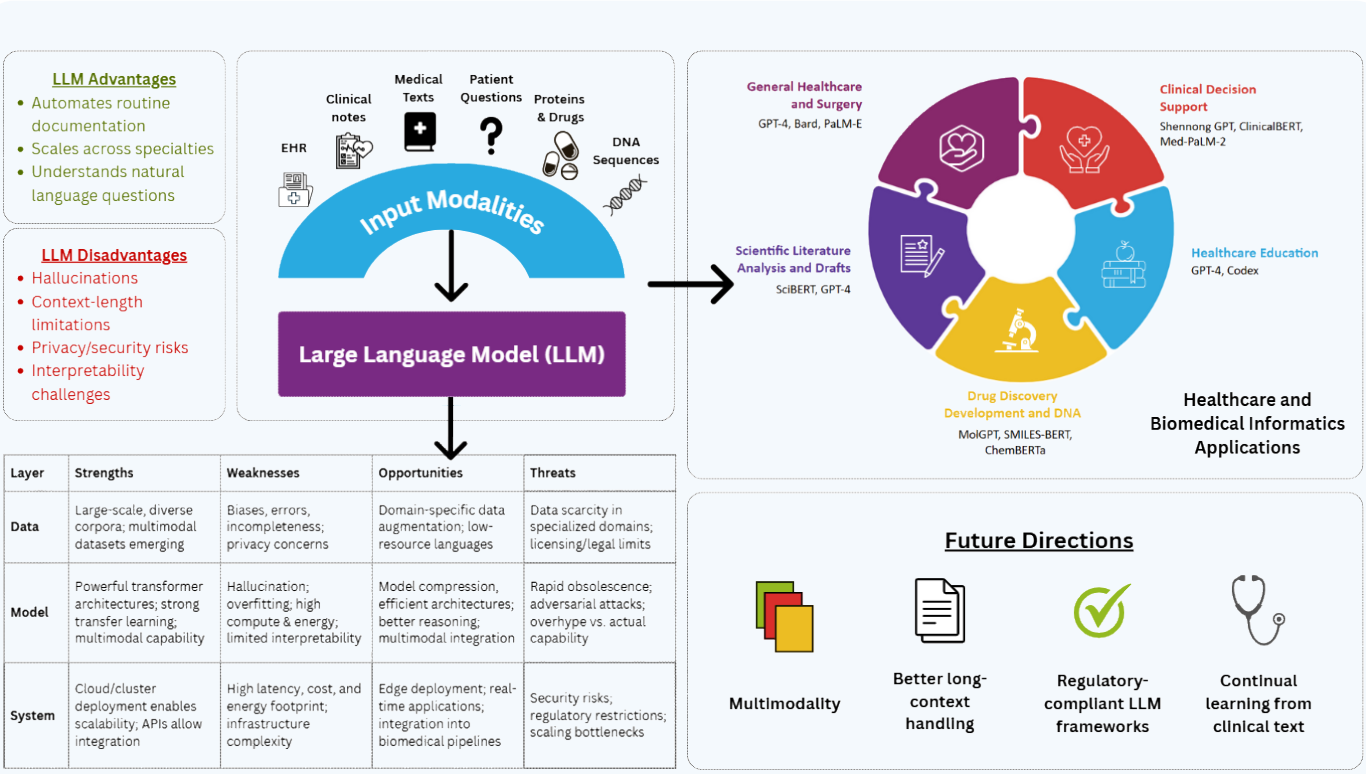
NER, relation extraction (RE), and question-answering (QA)[6]. BERT-based systems have since been widely applied, for example, generating radiology impressions[7], identifying patients with long COVID[8], and classifying clinical communications[9]. Building on this foundation, large language models (LLMs) such as ChatGPT[10], Gemini[11], and Claude[12], with billions of parameters, demonstrate strong capabilities in understanding, generation, summarization, translation, reasoning, and dialogue. They can perform zero- and few-shot learning, enabling generalization to unseen tasks without explicit fine-tuning. In healthcare, LLMs now support differential diagnosis from narratives, summarization of long records, patient–provider conversation simulation, biomedical code generation, literature review, drug development, and decision support[13,14]. However, despite their promise, the integration of LLMs into healthcare is not without challenges. Ethical, legal, and regulatory constraints, particularly around patient privacy, data security, and the reliability of artificial intelligence (AI)-generated outputs, necessitate careful evaluation. Challenges also include bias in training data, hallucination of medical facts, and model opacity, which can affect both patient safety and clinical decision-making. To provide a structured overview of LLM capabilities and challenges in healthcare, we organize our analysis using a SWOT framework (Strengths, Weaknesses, Opportunities, and Threats) across three layers: Data, Model, and System. **Figure 1** summarizes this SWOT analysis.

At the Data layer, LLMs benefit from large-scale, diverse corpora and emerging multimodal datasets, which support generalization across tasks. However, they remain vulnerable to biases, errors, incompleteness, and privacy concerns, while domain-specific data scarcity and legal or licensing restrictions pose ongoing threats. Opportunities include domain-focused data augmentation and expansion into low-resource languages. At the Model layer, transformer-based architectures enable powerful transfer learning, multimodal integration, and strong generalization. Weaknesses include hallucination, overfitting, high computational cost, and limited interpretability. Opportunities arise from model compression, efficiency improvements, and enhanced reasoning capabilities, whereas rapid obsolescence, adversarial attacks, and overhype represent potential threats.

At the System layer, cloud or cluster deployment and application programming interface (API)-based integration enable scalability and interoperability. Weaknesses involve high latency, infrastructure complexity, and energy cost, while opportunities include edge deployment, real-time applications, and integration into biomedical pipelines. Threats primarily relate to security risks, regulatory restrictions, and scaling bottlenecks. **Figure 2** provides an overview of the evolution of each layer.

Building on this structured SWOT framework, this article systematically reviews the current applications of LLMs across key areas in healthcare and biomedical informatics, highlighting both their potential and limitations. By integrating the insights from the SWOT analysis, we discuss concerns such as privacy, hallucination, and lack of rigorous validation, and outline future directions for the safe and effective adoption of LLMs in healthcare workflows:

- Clinical Decision Support and Patient Care
- Drug Discovery, Development, and DNA
- General Healthcare and Surgery



Layer	Strengths	Weaknesses	Opportunities	Threats
Data	Large-scale, diverse corpora; multimodal datasets emerging	Biases, errors, incompleteness; privacy concerns	Domain-specific data augmentation; low-resource languages	Data scarcity in specialized domains; licensing/legal limits
Model	Powerful transformer architectures; strong transfer learning; multimodal capability	Hallucination; overfitting; high compute & energy; limited interpretability	Model compression, efficient architectures; better reasoning; multimodal integration	Rapid obsolescence; adversarial attacks; overhype vs. actual capability
System	Cloud/cluster deployment enables scalability; APIs allow integration	High latency, cost, and energy footprint; infrastructure complexity	Edge deployment; real-time applications; integration into biomedical pipelines	Security risks; regulatory restrictions; scaling bottlenecks

Figure 1 Overview of LLM applications, strengths, limitations, and future directions in healthcare and biomedical informatics. Inputs include clinical notes, EHRs, medical texts, patient queries, and molecular data (proteins, drugs, and DNA). LLMs process these to support applications in general healthcare, bioinformatics, clinical decision-making, education, drug discovery, and literature analysis. Advantages include automation, scalability, and natural language understanding, while challenges include hallucinations, context limits, privacy, and interpretability. Adaptation strategies (pretraining, few-shot learning, instruction tuning, and retrieval-augmented generation) are compared for domain transfer, customization, and interpretability. Future directions emphasize multimodality, long-context handling, regulatory compliance, and continual learning from clinical text.

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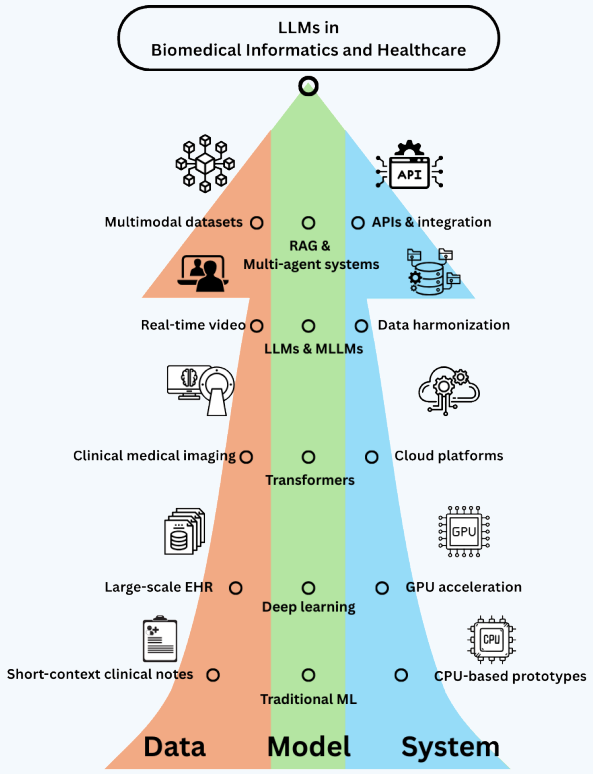


Figure 2 Overview of the LLM ecosystem in biomedical informatics and healthcare, spanning three interconnected layers: Data, Model, and System. These layers highlight how advances in data availability, model architectures, and system integration converge to enable retrieval-augmented generation, multi-agent systems, and real-world clinical applications.

- Healthcare Education
- Scientific Literature Analysis

Our work builds on and extends prior surveys of LLMs in healthcare and biomedical research. Compared to "Transformers and Large Language Models in Healthcare: A Review[15]," which primarily outlines transformer adoption across clinical NLP, imaging, EHR, and molecular applications, our survey emphasizes real-world deployment, performance evaluation, and integration into clinical and operational workflows. Relative to "The Ethics of ChatGPT in Medicine and Healthcare[16]," which systematically maps ethical and governance concerns, we combine both technical and ethical perspectives, highlighting practical challenges, data privacy, and explainability alongside actionable use cases. Finally, unlike "The Large Language Models on Biomedical Data Analysis: A Survey[17]," which focuses on genomic, proteomic, transcriptomic, radiomic, and single-cell analyses, our review covers a broader spectrum, including healthcare education, patient interaction, scientific literature synthesis, and drug discovery pipelines. Collectively, our survey highlights the transformative role of LLMs in healthcare and biomedical informatics, emphasizing both their potential to accelerate biomedical discovery and their challenges in clinical translation. The key messages of this work are threefold:

1) **Unifying data and knowledge:** LLMs can bridge structured (EHRs, omics) and unstructured (clinical notes, literature) data, enabling a more integrated view of biomedical information than was previously possible.
2) **Shaping trustworthy AI in medicine:** Their adoption is driving advances in interpretability, fairness, and regulation, which are essential to safely deploying AI in sensitive clinical contexts.
3) **Augmenting—not replacing—experts:** LLMs are already proving useful in tasks such as clinical decision support, drug discovery, and education, for example, summarizing complex patient histories to aid physicians in treatment planning. This highlights their role as transformative assistants that expand human capacity rather than substitutes for clinicians or researchers.

EVOLUTION OF LLMS IN HEALTHCARE

As illustrated in **Fig. 3**, the development of language models for healthcare has accelerated markedly since 2018. Early RNN and LSTM-based architectures have progressively given way to transformer-based foundation models, culminating in large-scale LLMs by 2023. This trajectory can be broadly characterized by three developmental phases: the LSTM era, the sequence-to-sequence (Seq2Seq) era, and the contemporary LLM era.

LSTM-based models

In the pre-LLM era, LSTM and RNN architectures were the predominant approaches for unimodal clinical text modeling. These models process sequences in a word-by-word manner, which constrained their ability to effectively model long-range contextual dependencies within clinical narratives. Qin et al. employed a Bi-directional LSTM (Bi-LSTM) framework combined with conditional random fields (CRF) for clinical NER, targeting the extraction of medical problems, tests, and treatments from deidentified discharge summaries and progress notes in the 2010 i2b2/VA Natural Language Processing Challenges for Clinical Records dataset. While the proposed method achieved an F1 score of 0.853, the authors reported persistent difficulty in accurately distinguishing medical problem entities, largely attributable to entity ambiguity and limited contextual representation[18].

Liu et al. extended the analysis to include the 2012 and 2014 i2b2 NLP challenge datasets, evaluating the comprehensive performance of LSTMs across multiple clinical NLP tasks. Their results indicated that LSTMs could surpass SOTA methods for concept extraction without requiring feature engineering. Specifically, the model achieved an F1 score of 0.8581 on the 2010 concept extraction dataset and F1 scores of 0.9229 and 0.9437 for clinical event detection and de-identification tasks, respectively. Despite these improvements, the authors observed that LSTM models failed to outperform SOTA methods for test event recognition in the 2012 dataset and ID instances in the 2014 dataset. Additionally, the models exhibited limitations in incorporating morphological information for fixed-size numerical tokens, highlighting persistent representational constraints inherent to recurrent architectures[19].

Rise of transformer-based models

To address the limitations of LSTM- and RNN-based models, transformer-based models with domain-specific pretraining were introduced. These transformer-based models primarily employed encoder-only architectures, such as BERT[5], with fewer models using an encoder–decoder architecture, such as the Text-to-Text Transfer Transformer (T5)[20]. The key innovation of these transformers is their use of self-attention, which enables both a deep understanding of contextual information and parallelized processing. These models can then be fine-tuned on domain-specific corpora, including biomedical text tailored for healthcare applications.

Some of the BERT-style models introduced during this period include BioBERT[6], BlueBERT[21], and ClinicalBERT[22]. These models follow the same pretraining approach as BERT, initially training on BooksCorpus and English Wikipedia, but incorporate additional pretraining on relevant domain-specific text. After pretraining, fine-tuning is performed for

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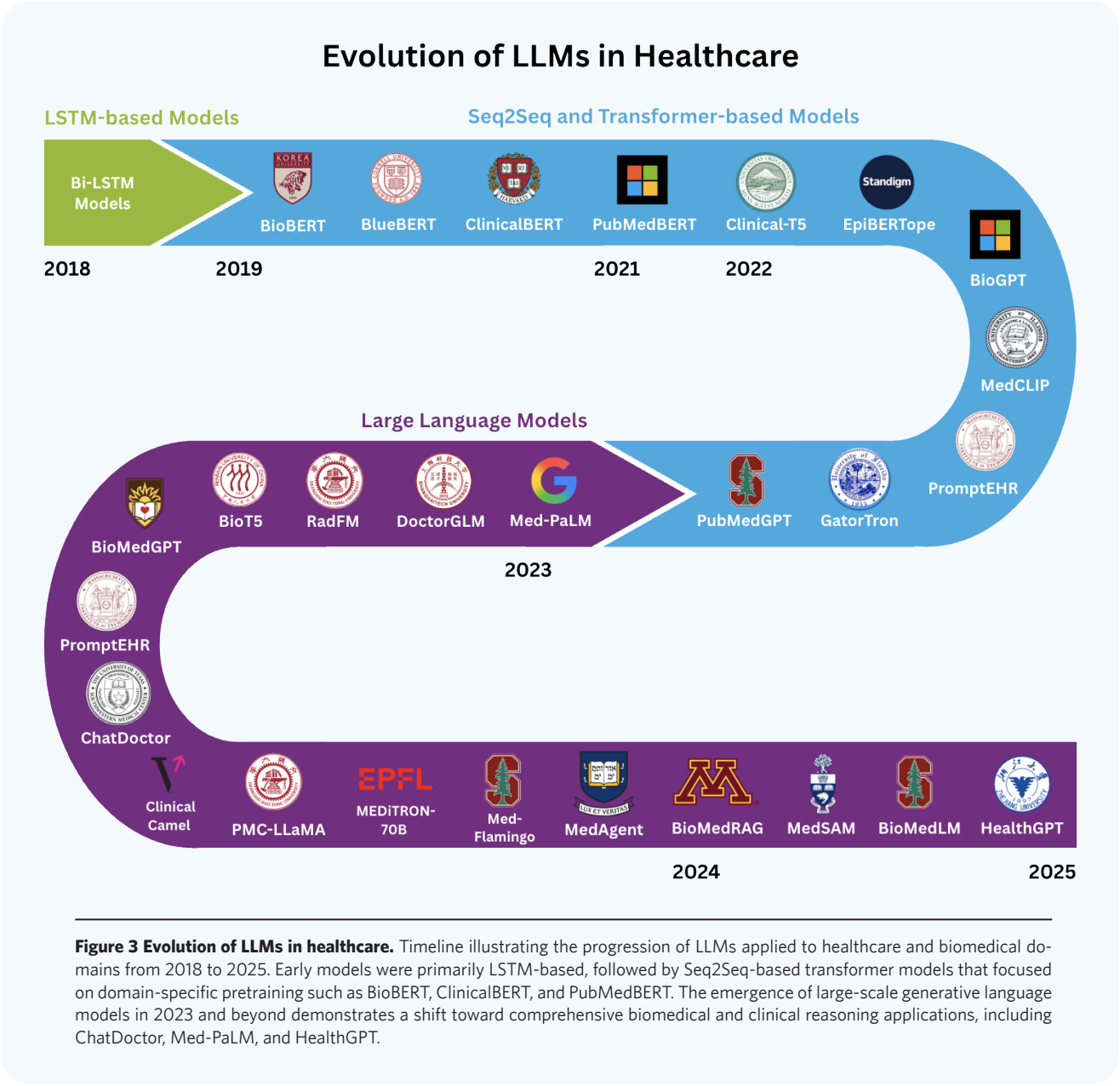


Figure 3 Evolution of LLMs in healthcare. Timeline illustrating the progression of LLMs applied to healthcare and biomedical domains from 2018 to 2025. Early models were primarily LSTM-based, followed by Seq2Seq-based transformer models that focused on domain-specific pretraining such as BioBERT, ClinicalBERT, and PubMedBERT. The emergence of large-scale generative language models in 2023 and beyond demonstrates a shift toward comprehensive biomedical and clinical reasoning applications, including ChatDoctor, Med-PaLM, and HealthGPT.

specific downstream tasks, such as NER and QA. These models excel in both token-level and sentence-level representations.

BioBERT is a key model focusing on biomedical text mining[6]. Using BERT as the base model, researchers continued pretraining on PubMed abstracts and PubMed Central (PMC) full-text articles. The model was then fine-tuned on biomedical NER, RE, and QA tasks. Evaluation was conducted on datasets covering diseases, chemistry, drug development, and gene-protein interactions, such as the National Center for Biotechnology Information (NCBI) disease corpus and BioCreative V Chemical Disease Relation (BC5DR) datasets[23,24]. BioBERT significantly outperformed previous SOTA models across all three biomedical text mining tasks, achieving an F1 score improvement of 0.62% for NER, 2.80% for RE, and a Mean Reciprocal Rank (MRR) improvement of 12.24% for QA through pretraining on large-scale biomedical corpora.

For BlueBERT, Peng et al. evaluated both BERT and ELMo[25] as baseline models on biomedicine tasks, including NER, RE, document classification, and inference. They found that BERT pretrained on PubMed abstracts and MIMIC-III clinical notes achieved the best results[26]. A key contribution of this work was the Biomedical Language Understanding Evaluation (BLUE) benchmark, designed to support the development of pretrained biomedical language representations. ClinicalBERT took a similar approach, fine-tuning base BERT on clinical notes to predict 30-day hospital readmission[27]. This model outperformed standard BERT in clinical text modeling and various baseline models on the readmission prediction task.

Encoder–decoder models, such as T5, also emerged during this period. Input text is processed by the encoder, which generates contextualized sequence representations, and the decoder, which produces output text autoregressively using cross-attention. An example in the healthcare domain is Clinical-T5, a T5 model pretrained on clinical text from the MIMIC-II dataset[28]. Its key tasks include document classification, NER, natural language inference (NLI), and real-world clinical tasks such as 30-day readmission and mortality risk prediction. Clinical-T5 significantly outperformed the general T5 model on these domain-specific tasks.

Following the success of domain-specific encoder and encoder–decoder models, research shifted toward leveraging transformer-based models for more complex and generative tasks, beyond simple discriminative tasks like classification and NER. This included multimodal data

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integration, paving the way for the LLMs that followed. Examples include PubMedBERT and BioGPT, both proposed by Microsoft Research[29,30]. PubMedBERT was pretrained solely on PubMed abstracts, without general pretraining on BooksCorpus and Wikipedia. This domain-specific pretraining led to superior performance compared to general BERT and SOTA healthcare-domain models such as BioBERT, SciBERT[31], and ClinicalBERT on NER, RE, semantic similarity, and other tasks. BioGPT followed a similar approach but used GPT as the base model, pretraining solely on PubMed articles to focus on generative tasks like text generation and QA.

In parallel, multimodal representation learning also advanced, exemplified by MedCLIP, which adapted the contrastive language-image pretraining (CLIP) framework to medical imaging, such as X-rays, paired with corresponding text reports[32]. This approach enabled zero-shot prediction and cross-modal retrieval, highlighting the potential of integrating diverse data modalities in biomedical AI. Alongside these multimodal developments, other GPT-based models emerged. BioMedLM, previously known as PubMedGPT, was released in 2022 and is autoregressively trained on PubMed abstracts and full articles[33]. Despite its smaller size (2.7B parameters), BioMedLM achieved competitive performance on multiple-choice biomedical QA tasks, scoring 69% on the Massive Multitask Language Understanding (MMLU) Medical Genetics Exam and 57.3% on Medical Multiple Choice Question Answering (MedMCQA), demonstrating the efficiency of domain-specific pretraining for generative biomedical tasks. Together, the success of these specialized transformer variants—particularly BioGPT with its generative capabilities and BioMedLM with its strong domain-specific performance, paved the way for larger, instruction-tuned foundation models, ultimately shaping the 2023 LLM revolution.

LLM revolution

Starting in 2023, there was an emergence of biomedical-focused LLMs built on general-purpose foundation models, such as GPT-3/4, Pathway Language Model (PaLM)[34], and Large Language Model Meta AI (LLaMA)[35]. The key shift in this era arose from the dramatic increase in model scale, with some models exceeding 50 billion parameters. In addition, instruction tuning and alignment techniques enabled these models to transition from general language predictors into clinically capable assistants, capable of complex reasoning and task-specific problem solving. Many LLMs developed during this period achieved superior performance on professional medical benchmarks, including standardized medical exams.

Several models demonstrated the ability of large-scale LLMs to capture human-level knowledge and reasoning when adapted to biomedical contexts. Google introduced Med-PaLM in 2023, a foundational model in the medical LLM revolution[36]. Med-PaLM was the first AI system to achieve a passing score of 60% on the Medical Question Answering (MedQA) dataset, which contains United States Medical Licensing Examination (USMLE) questions. Its successor, Med-PaLM 2, substantially improved performance, achieving 86.5% accuracy on the same dataset and outperforming physicians on long-answer consumer medical questions[37]. By leveraging the large-scale PaLM model as a backbone and performing task-specific fine-tuning on MultiMedQA, a curated collection of seven medical QA datasets, this work illustrated that scale combined with targeted fine-tuning enables advanced medical reasoning. Another model, BioMedGPT, employed self-supervised pretraining on diverse biomedical datasets and is capable of processing both imaging and textual data[38]. This multimodal pretraining allowed BioMedGPT to achieve SOTA performance in radiology report generation and MedQA while remaining computationally more efficient than larger, closed-source alternatives.

Following these early breakthroughs, the field shifted toward leveraging open-source foundation models. After Meta released LLaMA, several research groups developed fine-tuned biomedical models to match or exceed the performance of closed-source systems. For example, PMC-LLaMA and Clinical Camel applied instruction tuning and incorporated knowledge from PMC to improve clinical reasoning tasks[39,40]. This trend continued with MEDITRON-70B, a large-scale open-weight model specifically pretrained on PubMed abstracts, full-text articles, and clinical guidelines[41].

During this period, proprietary efforts also focused on specialized clinical systems. GatorTron leveraged massive EHR datasets for clinical decision support, while ChatDoctor and DoctorGLM explored refinement of conversational interfaces for patient-facing interactions[42–44]. These models exemplify the transition from simple information retrieval systems to interactive clinical assistants capable of reasoning, summarization, and guidance. The most recent wave, from 2023 to 2025, marked a shift toward multimodality, retrieval-augmented generation (RAG), and agentic architectures. Building on early models like MedCLIP, newer architectures such as Med-Flamingo and MedSAM integrate visual and textual reasoning at scale. MedSAM, in particular, adapts the Segment Anything Model (SAM)[45] for medical imaging, demonstrating how LLM-inspired architectures can generalize across diverse image segmentation tasks[46,47]. To address limitations in context integration and reduce hallucinations, RAG-based models like BioMedRAG incorporate external knowledge bases, including General BioMedical and Complementary and Integrative Health Triples (GIT), Chemistry-Protein (CHEMPROT), and Drug–Drug Interaction (DDI) datasets[48]. In parallel, agentic architectures such as MedAgent and HealthGPT implement multi-step reasoning, tool use, and long-term patient health management, marking a transition from task-specific models to comprehensive, inter-active healthcare platforms[49,50]. A summary of selected models is shown in Table 1.

HOW LARGE LANGUAGE MODELS WORK?

LLMs are DL systems designed to process and generate human language. They are built on the transformer architecture, which enables parallel processing of entire sequences and captures both short- and long-range dependencies efficiently. Although their core training objective is simple, predicting the next token given the preceding context, scaling this principle to massive datasets and model sizes yields powerful emergent capabilities such as dialogue, summarization, and code generation. An overview of how transformer-based LLMs work is shown in **Fig. 4**.

Architecture overview

The transformer architecture consists of two main stages:

1) **Input Stage:** Prepares raw text for the model by converting it into a numerical representation with positional information.
2) **Transformer Stage:** Uses stacked self-attention and feed-forward layers to build contextualized token representations, which can then be used for generation or other tasks.

1) Input Stage: Raw text is first tokenized into smaller units called tokens using methods such as WordPiece or SentencePiece. For example, "Mutations in the gene BRCA2 can lead to breast cancer" may be split into ("Mutations," "in," "the," "gene," "BRCA2," "can," "lead," "to," "breast," "cancer"]. Each token is mapped to an integer ID in the model's vocabulary and then to a high-dimensional embedding vector:

$$x_i \in R^d$$

where x_i is the embedding for the i-th token and d is the embedding dimension.

To encode the order of tokens, a positional encoding vector PE_i is added to each embedding:

$$z_i = x_i + \mathrm{PE}_i$$

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Table 1 Selected biomedical and clinical large language model summary.

Model Name	Base Model	Year	Domain	Short Description	Model Size
BioBERT[6]	BERT	2019	Biomedical	Pretrained on PubMed and PMC for biomedical text mining	110M
BlueBERT[21]	BERT	2019	Biomedical, Clinical	Biomedical and clinical BERT trained on PubMed and MIMIC-III notes	Base: 110M; Large: 340M
SciBERT[31]	BERT	2019	Biomedical, Computer Science	SciBERT is a BERT-based model pre-trained on full text scientific papers. It improves performance on scientific NLP tasks such as entity recognition and relation classification.	~110M parameters
ClinicalBERT[22]	BERT	2019	Clinical	Pretrained on MIMIC-III clinical notes and fine-tuned for downstream clinical tasks	110M
PubMedBERT[29]	BERT	2021	Biomedical	Pretrained from scratch on PubMed abstracts using in-domain vocabulary	110M
BioGPT[30]	GPT-2	2022	Biomedical	Generative transformer trained on 15M PubMed abstracts; strong biomedical NLP performance	347M
Clinical-T5[28]	T5	2022	Clinical	Domain-adapted generative model was further pretrained on MIMIC-III clinical notes	Base: ~220M; Large: ~770M
EpiBERTope[51]	BERT	2022	Genomics, Immunology	Sequence-based BERT model trained on Swiss-Prot for epitope prediction	12 layers, 252 hidden units
GatorTron[42]	Megatron-BERT	2022	Clinical	Large clinical language models trained on >90B words, including UF Health notes and PubMed	345M–8.9B
PromptEHR[52]	BART	2022	EHR, Clinical	Prompt-based model for generating synthetic longitudinal EHRs with privacy evaluation	N/A
MedCLIP[32]	CLIP	2022	Radiology	Contrastive learning on unpaired medical images and clinical reports	N/A
PubMedGPT[53]	GPT-2	2022	Biomedical	GPT-style model trained exclusively on PubMed abstracts and papers	2.7B
BioT5[54]	T5	2023	Biomedical, Chemical, Protein	Cross-modal model pretrained on molecules, proteins, and biomedical text	252M
Med-PaLM[36]	Flan-PaLM	2023	Clinical	Instruction-tuned large language model aligned to the medical domain	540B
ChatDoctor[43]	LLaMA	2023	Clinical	Fine-tuned using 100K real-world patient–doctor dialogues	7B
Clinical Camel[39]	LLaMA-2	2023	Clinical	QLoRA-fine-tuned LLaMA-2 optimized for clinical QA and note generation	13B, 70B
Med-PaLM 2[37]	PaLM-2	2023	Clinical	Expert-level instruction-tuned medical QA model	N/A
PMC-LLaMA[40]	LLaMA	2023	Biomedical, Clinical	Open-source medical LLM trained on biomedical papers and textbooks	113B
RadFM[55]	MedLLaMA-13B	2023	Radiology	Generalist radiology foundation model for 2D/3D multimodal reasoning	14B
BioMedGPT[38]	Transformer	2024	Biomedical	Open-source biomedical vision-language generalist model	33M–182M
BioMedLM[33]	GPT-style	2024	Biomedical	GPT-style model trained solely on PubMed abstracts and full texts	2.7B
MedSAM[46]	SAM	2024	Medical Imaging	Foundation model for promptable medical image segmentation	~94M
Me-LLaMA[56]	LLaMA-2	2024	Biomedical, Clinical	Open-source medical LLM family trained on biomedical and clinical corpora	13B, 70B
MedAgent[49]	GPT	2024	Clinical	Multi-agent framework for zero-shot clinical reasoning	N/A
MiniGPT-Med[57]	EVA, LLaMA-2 7B	2024	Medical Imaging	Multimodal model for radiology VQA and report generation	~7B
BiomedRAG[48]	LLaMA2-13B, MedLLaMA-13B, GPT-4	2025	Biomedical	Retrieval-augmented generation framework for biomedical NLP	13B
HealthGPT[50]	Phi-3-mini, Phi-4	2025	Medical	Unified vision-language medical model	3.8B–14B

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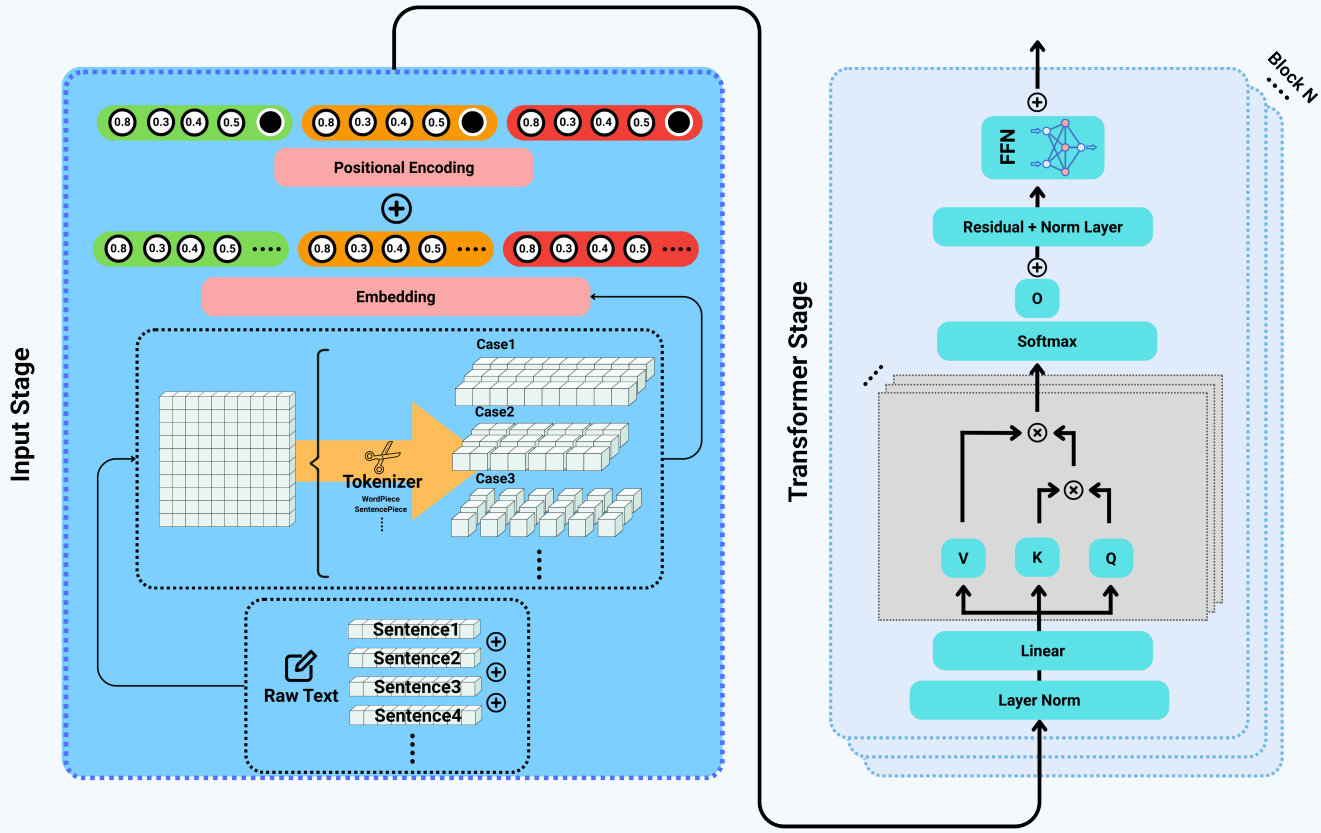


Figure 4 Overview of a transformer-based LLM workflow. The **Input Stage** begins with raw text tokenization into discrete tokens, which are mapped to dense vector embeddings. Positional encodings are added to capture word order. In the **Transformer Stage**, the sequence passes through multiple stacked attention blocks, each consisting of layer normalization, multi-head self-attention, residual connections, and feed-forward networks. This iterative process builds rich contextual representations of the input text.

Positional encodings may be fixed (e.g., sinusoidal) or learned; models like BERT and GPT typically use learned encodings. The resulting sequence of z_i vectors is passed to the transformer blocks.

2) Transformer Stage: The transformer processes the sequence through N identical blocks, each containing:

1) **Layer Normalization:** Normalizes inputs to stabilize and speed up training:

$$\text{LayerNorm}(z_i) = \frac{z_i - \mu(z_i)}{\sigma(z_i)}$$

2) **Multi-Head Self-Attention:** Projects inputs into queries (Q), keys (K), and values (V):

$$Q = ZW^Q, \quad K = ZW^K, \quad V = ZW^V$$

Attention weights are computed as

$$\text{Attention}(Q,K,V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}V\right)$$

Multiple attention heads run in parallel: $\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W^o$

This allows the model to learn relationships from different representation subspaces.

3) **Residual Connection + Layer Norm:** The attention output is added to the block input (residual connection) and normalized again.
4) **Feed-Forward Network (FFN):** Applies two linear transformations with a non-linear activation (e.g., ReLU or GELU):

$$\text{FFN}(x) = W_2 \cdot \sigma(W_1 x + b_1) + b_2$$

Another residual connection and layer normalization follow.

Stacking these blocks allows the model to construct deep contextual representations, capturing both local and global dependencies in text.

Output stage

After the final transformer block, a layer normalization is applied. The resulting hidden states are projected to the vocabulary dimension and passed through a Softmax function to produce a probability distribution over the next token:

$$\hat{y} = \text{softmax}(h_{\text{final}}W^T + b)$$

For each generation, the most probable token (or a token sampled according to the distribution) is appended to the sequence, and the process repeats.

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LLM ENGINEERING

Recent advances in LLMs have been driven not only by increases in model scale and data but also by innovations in LLM engineering, the set of techniques that govern how models are prompted, augmented with external knowledge, and organized into more complex systems. In healthcare and biomedical engineering, these methods are particularly important because clinical tasks often require reliable reasoning, domain grounding, transparency, and adaptability to evolving knowledge. **Figures 5 and 6** provide an overview of core LLM engineering paradigms discussed in this section. The first figure summarizes foundational techniques such as prompting strategies, reasoning frameworks, and RAG, which enhance model performance without parameter updates. The second figure illustrates higher-level agentic and system-based designs that combine LLMs with tools, memory, and decision-making modules to support more complex, multi-step clinical workflows. Together, these approaches form the engineering backbone for deploying LLMs in real-world healthcare applications.

1) Zero-Shot/Few-Shot Prompting, Chain/Tree-of-Thoughts: Few-shot learning refers to solving a new task from only a small number of examples. In large pretrained language models, this is typically implemented as *in-context learning*: at inference time, we place a natural-language instruction and a few input–output demonstrations directly in the prompt, and the model performs the task by continuing the text, without any gradient updates or fine-tuning. Following this setup, it is common to distinguish *zero-shot* (instruction only), *one-shot* (a single demonstration), and *few-shot* (multiple demonstrations, often as many as fit in the context window) evaluation regimes. As illustrated in our **Fig. 5**, the prompt functions as a compact task specification (instruction + examples + query), and scaling up the model tends to improve how effectively it can use these in-context examples, often yielding larger gains in few-shot performance than in zero-shot performance on many benchmarks[58].

Chain-of-thought (CoT) prompting is a simple prompting method that improves reasoning in LLMs by asking the model to generate intermediate natural language steps before the final answer[59]. Instead of giving only input–output exemplars, the prompt includes triples of *(input, chain of thought, and output)*. This approach can significantly improve performance on arithmetic, commonsense, and symbolic reasoning tasks, with especially large gains on challenging multi-step problems such as Grade School Math 8K (GSM8K). A key finding is that the benefit is an emergent property of model scale: small models often produce fluent but illogical chains of thought, while models around 100B parameters and

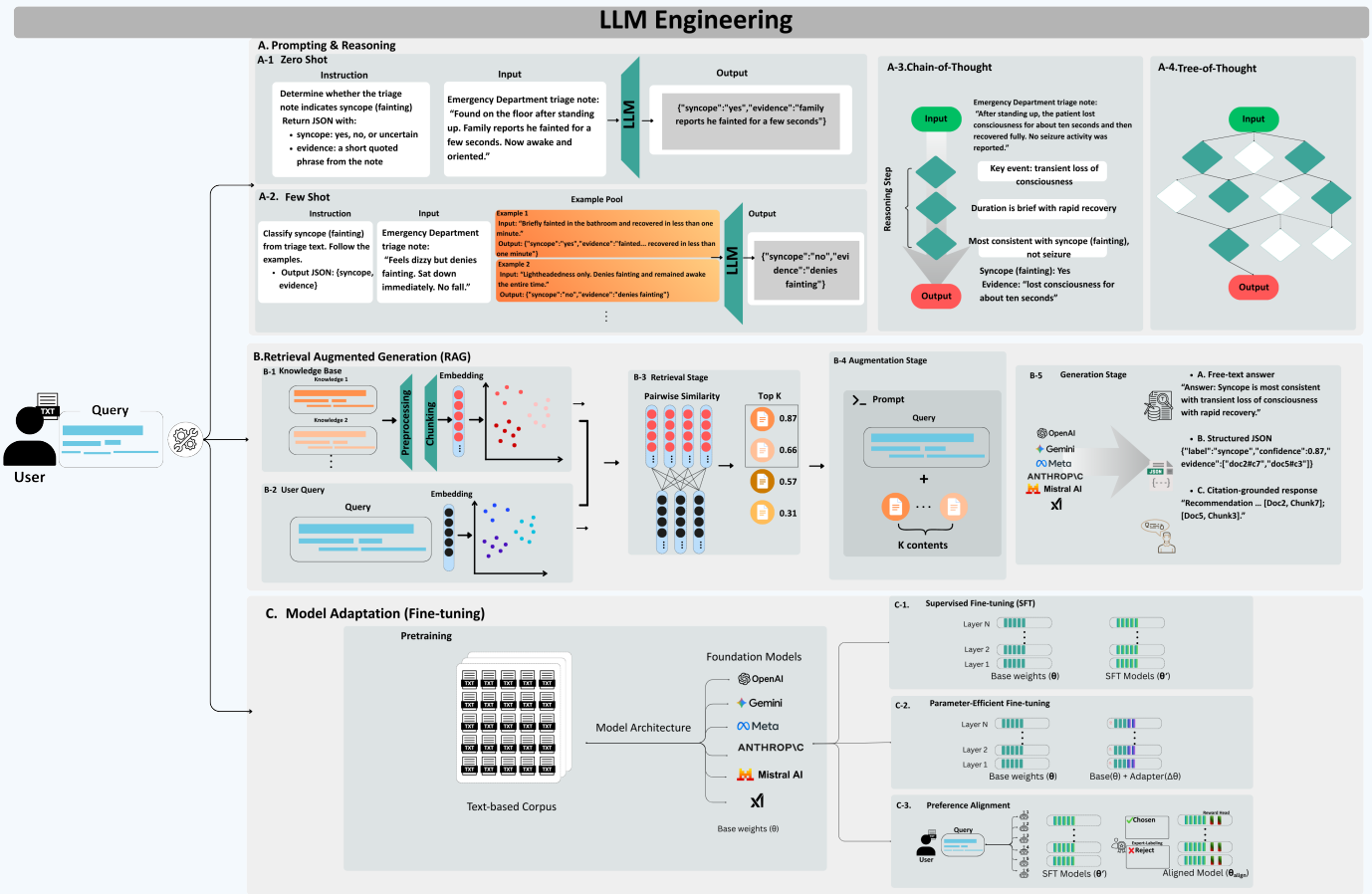


Figure 5 Overview of LLM engineering techniques for healthcare applications. This figure summarizes the core components of LLM engineering, organized into three layers: (A) prompting and reasoning, including zero-shot and few-shot prompting, chain-of-thought, and tree-of-thought strategies for structured clinical reasoning; (B) retrieval-augmented generation (RAG), illustrating knowledge indexing, embedding, retrieval, and evidence-grounded generation to improve factuality, transparency, and updatability; and (C) model adaptation, covering supervised fine-tuning, parameter-efficient fine-tuning, and preference alignment built on foundation models. Together, these techniques enable reliable, interpretable, and domain-adapted deployment of LLMs in healthcare and biomedical engineering workflows.

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larger show strong improvements. Ablations suggest the gains are not explained by writing equations alone, adding extra tokens as "variable compute," or providing reasoning after the answer; the sequential reasoning steps appear to matter. The method is also relatively robust across different annotators and exemplar sets, and it can support out-of-domain length generalization in symbolic tasks.

Tree of thoughts (ToT) is an inference framework that helps a language model do more deliberate problem solving[60]. In standard inference, the model makes token-level decisions in a left-to-right way, so it can fall short on tasks that need exploration, strategic lookahead, or backtracking. ToT generalizes CoT prompting by exploring a ToTs, where each thought is a coherent unit of text that serves as an intermediate step toward problem solving. In ToT, each node is a state that represents a partial solution, and the model generates multiple candidate thoughts for the next step. Then the model self-evaluates these candidates and uses a search algorithm, such as breadth-first search or depth-first search, to decide which states to keep exploring. Because ToT keeps diverse alternatives and can look ahead or backtrack, it supports deliberate decision-making instead of committing to one early choice. In the experiments (Game of 24, Creative Writing, and Mini Crosswords), ToT improves problem-solving performance compared to CoT prompting.

2) Retrieval-Augmented Generation: RAG is a practical way to help LLMs handle knowledge-intensive tasks. Instead of relying only on facts stored in model parameters, a RAG system first retrieves relevant passages from an external knowledge source and then generates an answer that is grounded in those passages. This approach is useful because it can improve factual accuracy and reduce hallucinations, and it can also make outputs easier to verify by linking them to the retrieved evidence. Another benefit is that the system can be updated by changing the retrieval index or the knowledge base without fully retraining the language model[61]. More recently, survey work has shown RAG as a broader method for LLMs and summarized common design choices such as indexing, retrieval strategies, and how retrieved text is integrated during generation. These developments have made RAG a widely used baseline for building more reliable and knowledge-aware NLP systems[62].

RAG can contribute to healthcare in three perspectives: equity, reliability, and personalization. For equity, the system can retrieve medical literature, clinical guidelines, and case reports that are relevant to different subpopulations so it can support bias reduction and help address disparity mitigation. For reliability, the retrieved evidence can support mistake alleviation by reducing incorrect content and it can support transparency enhancement by making the sources more traceable for verification. For personalization, the system can connect external knowledge with patient records and clinical data to support health management, and it can help precision medicine by using individualized clinical and molecular data to guide more tailored recommendations[63].

3) Model Adaptation and Fine-tuning: While prompting and retrieval-based methods enable strong performance without modifying model parameters, many healthcare applications benefit from *model adaptation*, where a pretrained foundation model is further optimized for

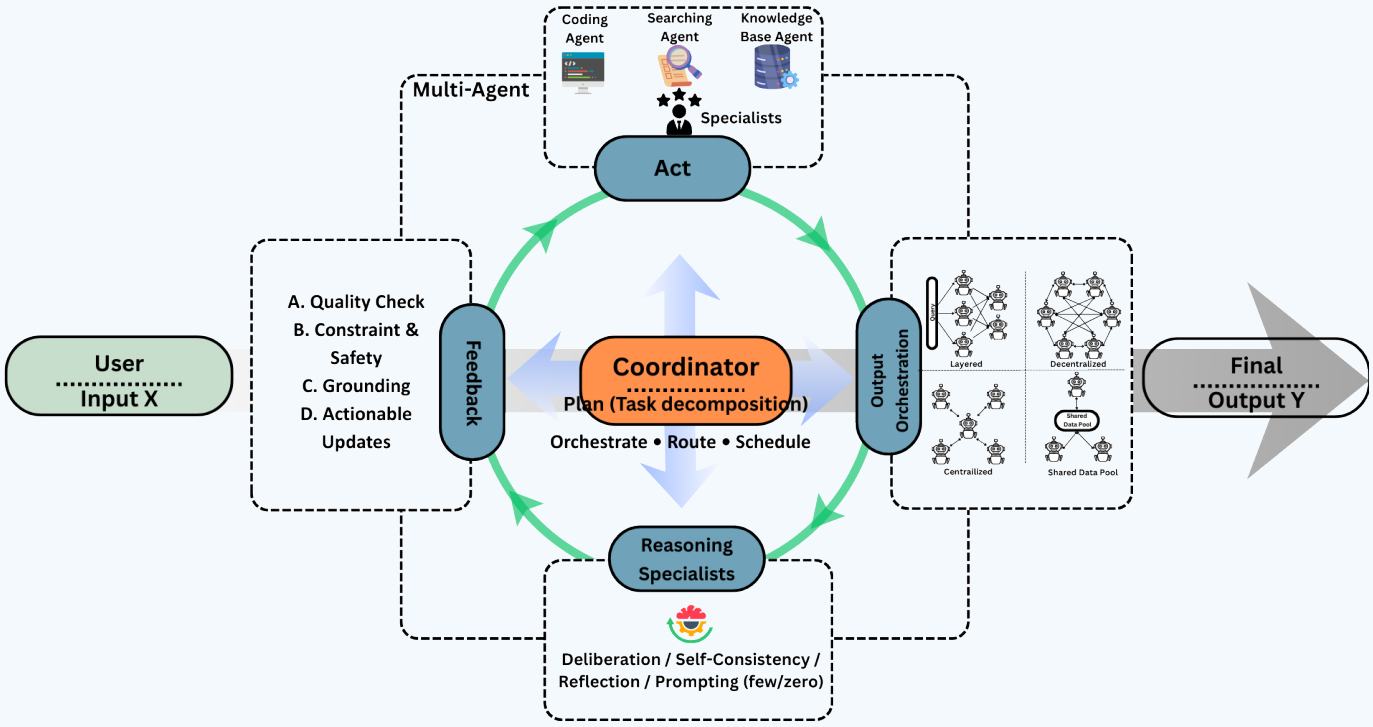


Figure 6 Agentic LLM system architecture for healthcare applications. This figure illustrates a coordinator-centered, multi-agent LLM framework in which a central planning module decomposes user inputs into subtasks and orchestrates their execution across specialized agents. Reasoning specialists support deliberation, self-consistency, reflection, and adaptive prompting, while action-oriented agents (e.g., coding, search, and knowledge-base agents) perform task-specific operations. Output orchestration integrates results from layered, centralized, decentralized, or shared-data-pool agent configurations. A feedback loop enforces quality checks, safety and constraint compliance, grounding to evidence, and actionable updates before producing the final output. This architecture enables scalable, transparent, and reliable multi-step reasoning for complex healthcare and biomedical engineering workflows.

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Table 2 LLM adaptation methods in healthcare.

Method	Data Required	Compute Cost	Typical Use	Risks
Zero-shot	None	Low	Rapid prototyping	Hallucinations
Few-shot	Prompt examples	Low	Rare diseases	Prompt sensitivity
Full fine-tuning	Large labeled data	Very high	Production deployment	Overfitting
LoRA/PEFT	Small datasets	Medium	Institutional models	Capacity limits
RLHF	Expert feedback	Very high	Safety-critical tasks	Bias amplification

domain-specific tasks. Model adaptation typically builds on large general-purpose or biomedical foundation models and aims to improve clinical accuracy, robustness, and alignment with expert expectations.

Supervised fine-tuning (SFT) is the most direct form of adaptation, in which models are trained on curated datasets of task-specific input–output pairs, such as clinical note classification, MedQA, or report generation. In healthcare, SFT datasets often incorporate expert-annotated labels or clinician-authored responses, enabling models to better capture domain terminology, documentation styles, and clinical decision patterns. However, full fine-tuning requires substantial computational resources and raises concerns related to data privacy, reproducibility, and model overfitting when labeled clinical data are limited.

To address these challenges, parameter-efficient fine-tuning (PEFT) methods have gained traction. Techniques such as low-rank adaptation (LoRA), adapters, and prefix tuning modify only a small subset of parameters while keeping the base model weights fixed. PEFT approaches significantly reduce memory and computational costs, making them particularly attractive for healthcare settings where resources are constrained and model updates must be frequent. Moreover, PEFT enables easier model sharing and deployment across institutions without exposing or retraining full model weights.

Beyond task performance, preference alignment methods aim to align model outputs with human values and domain-specific expectations. Approaches such as reinforcement learning from human feedback (RLHF) or AI-generated feedback (RLAIF) refine model behavior by incorporating judgments about correctness, safety, and usefulness. In clinical contexts, alignment is critical for reducing harmful or misleading outputs, improving response calibration, and encouraging transparency in reasoning and uncertainty expression. While alignment techniques are still an active area of research in healthcare, they represent an important step toward trustworthy and clinically deployable LLM systems.

Together, SFT, parameter-efficient adaptation, and preference alignment form a complementary toolkit for customizing foundation models to healthcare and biomedical engineering tasks. These approaches trade off flexibility, cost, and data requirements and are often combined with prompting and retrieval-based techniques to achieve optimal performance in real-world clinical workflows. A summary of these methods is shown in Table 2.

Agent-based and multi-agent systems

Beyond single-prompt interactions, recent LLM systems increasingly adopt agent-based architectures that decompose complex tasks into coordinated sequences of reasoning, tool use, and verification steps. In these systems, one or more LLM-powered agents interact through explicit roles, control loops, and feedback mechanisms, enabling more reliable and scalable problem solving. Agent-based designs are particularly relevant in healthcare and biomedical engineering, where tasks such as clinical decision support, literature synthesis, and patient-specific reasoning often require multi-step workflows, external tools, and iterative validation.

A common design pattern is the use of a central coordinator or planner that performs task decomposition and orchestration. Given a user query, the coordinator identifies subtasks, routes them to specialized agents, and manages execution order and dependencies. These specialized agents may include reasoning agents, retrieval or search agents, coding or data-processing agents, and knowledge-base agents, each optimized for a specific function. This modular structure allows systems to combine heterogeneous capabilities while maintaining interpretability and control over intermediate steps.

Reasoning specialists play a key role in agentic systems by performing deliberation, self-consistency checks, and reflective reasoning. Techniques such as debate, self-reflection, and majority voting across multiple reasoning paths can be used to reduce brittleness and improve robustness, especially for high-stakes clinical tasks. These agents often leverage prompting strategies such as few-shot or zero-shot reasoning and may iteratively refine intermediate outputs before execution or reporting.

The action and tool-use layer enables agents to interact with external resources, including databases, clinical guidelines, calculators, code interpreters, and EHR interfaces. Action-oriented agents translate model outputs into executable commands, while safeguarding mechanisms enforce constraints related to safety, privacy, and scope of operation. In healthcare settings, this separation between reasoning and action is critical for ensuring that recommendations are grounded, auditable, and compliant with clinical workflows.

Feedback and quality-control loops are another defining feature of agent systems. Outputs generated by individual agents may be evaluated against predefined criteria such as factual consistency, safety constraints, grounding in evidence, and actionability. Feedback can be used to trigger revisions, escalate tasks to additional agents, or halt execution when uncertainty is high. These mechanisms support transparency and enable systems to surface confidence estimates or justifications alongside final outputs.

Finally, agent-based systems can be organized under different coordination topologies, including centralized, decentralized, layered, or shared-memory architectures. Centralized designs simplify control and auditing, while decentralized or multiagent configurations can improve scalability and fault tolerance. In biomedical applications, the choice of coordination strategy reflects trade-offs between interpretability, computational cost, and system autonomy.

Overall, agent-based and multi-agent LLM systems represent a shift from static text generation toward dynamic, goal-driven computation. By integrating planning, reasoning, tool use, and feedback, these architectures provide a flexible foundation for building trustworthy and extensible LLM-powered healthcare systems.

METHODOLOGY

We conducted a systematic review using a multi-stage search strategy to capture research on LLMs in healthcare and biomedical informatics. Searches were performed in five major databases—ACM Digital Library, IEEE Xplore, PubMed, ScienceDirect, and Scopus—covering computational, biomedical, and clinical informatics perspectives. To expand coverage, we also applied backward and forward citation tracking.

Search strings combined "Large Language Model" with domain-specific terms such as "Bioinformatics," "Genomics," "Healthcare," "Clinical Decision Support," "Electronic Health Records," and "Biomedical NLP," as shown in Table 3. Eligible studies were written in English, focused on LLM use, evaluation, or development in biomedical or

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Table 3 Search query and results in ACM, IEEE Xplore, PubMed, ScienceDirect, and Scopus.

Academic Database	Search Query	Results
ACM	[All: "large language model"] AND ([[All: bioinformatics] OR [All: genomics] OR [All: "healthcare analysis"] OR [All: "clinical decision support"] OR [All: diagnosis] OR [All: therapy] OR [All: "drug discovery"] OR [All: dna] OR [All: surgery] OR [All: "healthcare education"] OR [All: "hospital operations"] OR [All: "scientific literature"] OR [All: "biomedical nlp"]]) AND [E-Publication Date: (01/01/2023 TO 05/31/2025)]	598
IEEE	(("large language model") AND (bioinformatics OR genomics OR "healthcare analysis" OR "clinical decision support" OR diagnosis OR therapy OR "drug discovery" OR dna OR surgery OR "healthcare education" OR "hospital operations" OR "scientific literature" OR "biomedical nlp")) Range 2023–2025	233
PubMed	"large language model"[All Fields] AND (bioinformatics[All Fields] OR genomics[All Fields] OR "healthcare analysis"[All Fields] OR "clinical decision support"[All Fields] OR diagnosis[All Fields] OR therapy [All Fields] OR "drug discovery"[All Fields] OR DNA[All Fields] OR surgery[All Fields] OR "healthcare education"[All Fields] OR "hospital operations"[All Fields] OR "scientific literature"[All Fields] OR "biomedical NLP"[All Fields])	1,146
ScienceDirect	"large language model" AND (bioinformatics OR genomics OR "healthcare analysis" OR "clinical decision support" OR diagnosis OR therapy OR "drug discovery") (2023–2025 Research Articles, Title, abstract, or author-specified keywords)	326
Scopus	"large language model" AND (DNA OR surgery OR "healthcare education" OR "hospital operations" OR "scientific literature" OR "biomedical NLP") (2023–2025 Research Articles, Title, abstract, or author-specified keywords)	135

healthcare contexts, and available in full text. Filters were applied when possible to prioritize recent, research-focused publications. Study selection followed the PRISMA framework[64]. **Figure 7** summarizes retrieval, screening, and exclusion steps, including removal for duplication, lack of domain relevance, or insufficient technical detail. Alongside peer-reviewed studies, we reviewed technical artifacts central to LLM use in biomedicine, including model documentation, API references, deployment guides, and fine-tuning resources from proprietary platforms and open-source ecosystems.

Finally, included works were thematically categorized by application area (e.g., bioinformatics, clinical decision support, literature analysis), methodological approach (e.g., pretraining, prompt engineering, domain adaptation), evaluation strategy, and reported challenges (e.g., fairness, interpretability, workflow integration). This process distilled current patterns and future directions for LLMs in healthcare and biomedical informatics.

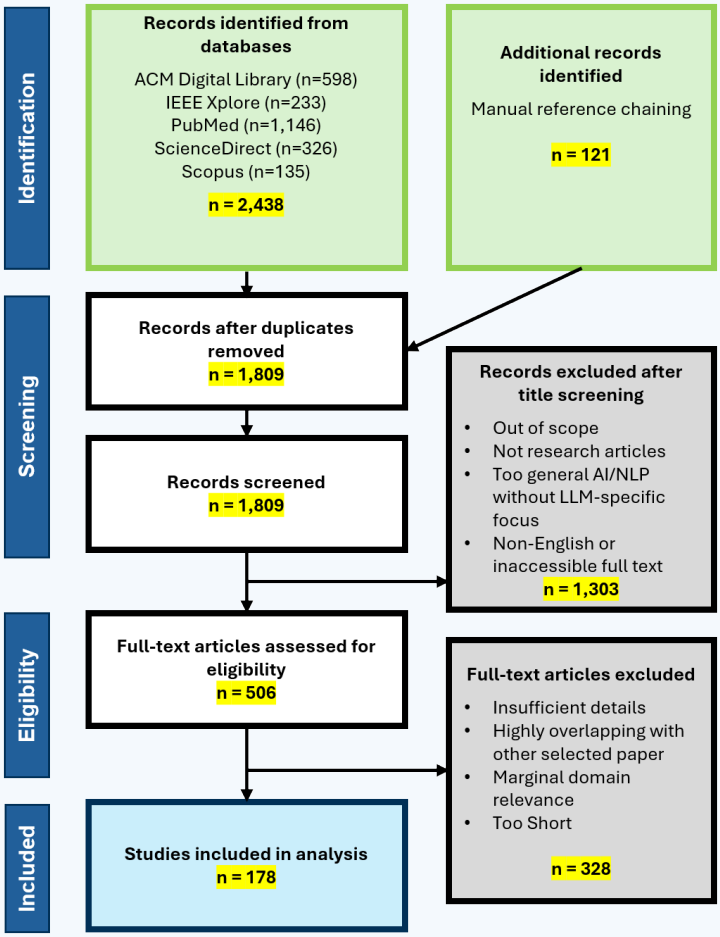


Figure 7 PRISMA flow diagram for relevant literature search and assessment.

APPLICATIONS

LLMs are being applied across a wide spectrum of healthcare and biomedical informatics domains, with use cases that span from research to patient-facing care. We group these applications into five major areas: clinical decision support and patient care, drug discovery, development, and DNA; general healthcare and surgery, healthcare education, and scientific literature analysis. Within these areas, LLMs are already demonstrating potential impact across diverse diseases and conditions. For example, they have been explored in diabetes management and lifestyle-related disorders such as obesity, applied to patient communication in cardiac arrest and education around stroke contraindications, and used to support mental health care for major depressive disorder. Specialty applications, such as in otolaryngology—head and neck surgery, further highlight their breadth. Together, these examples illustrate that LLMs

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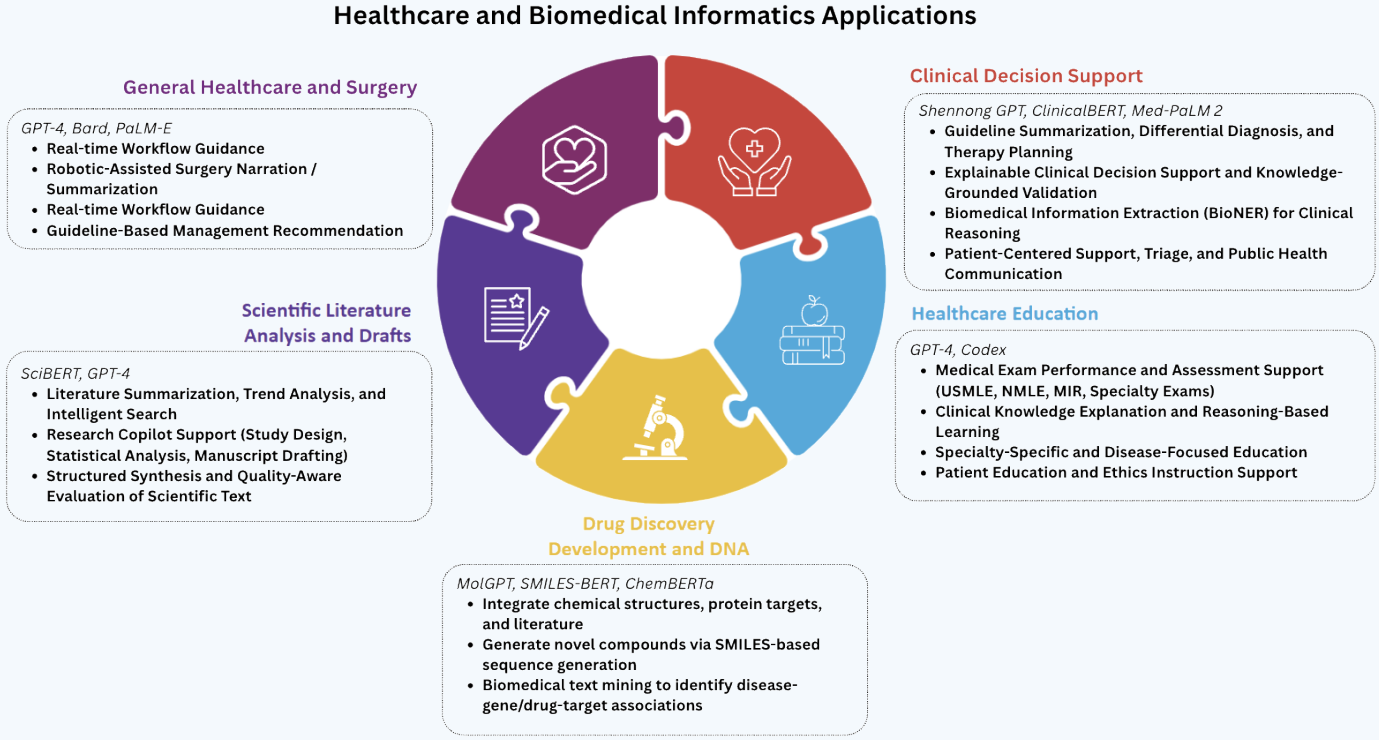


Figure 8 Applications of LLMs in Healthcare and Biomedical Informatics. The diagram illustrates key domains where LLMs are being applied, including General Healthcare and Surgery, Clinical Decision Support, Healthcare Education, Drug Discovery and DNA Research, and Scientific Literature Analysis. Each domain highlights representative models and specific tasks, such as real-time workflow guidance, differential diagnosis support, exam performance assistance, novel compound generation, and literature summarization.

are not confined to generic tasks but are being adapted to disease-specific contexts, opening opportunities to improve diagnosis, treatment, patient education, and biomedical discovery. Figure 8 provides an overview of the key application areas in healthcare and biomedical informatics where LLMs are being applied.

Clinical decision support and patient care

LLMs are increasingly applied to clinical decision support and patient care, assisting with data interpretation, guideline summarization, differential diagnosis, and therapy planning. Their ability to understand clinical language and reason over medical knowledge enables real-time decision support, personalized insights, and enhanced provider–patient communication[65–67].

Yang et al. leveraged GPT-3 in a literature-based explainable AI system to support clinician validation of AI-generated recommendations, promoting trust through evidence rather than model transparency. While promising across three disease areas, GPT-3's limitations in processing biomedical texts remain critical in high-stakes settings[68]. Lin et al. evaluated multiple LLMs in opioid medication adjustment for palliative care, showing potential utility but emphasizing caution and the need for workflow integration research[69]. Similarly, Rajashekar et al. found that GutGPT enhanced usability and trust in managing upper gastrointestinal bleeding, with clinicians using it to validate intuition[70].

Bian et al. developed SKIPNER, a one-shot biomedical named entity recognition (BioNER) framework that enhances LLM performance in domain-specific information extraction tasks. Drawing inspiration from COT prompting, SKIPNER decomposes BioNER into two subtasks: span extraction and entity type determination. The first step uses in-context one-shot learning for span detection, while the second step incorporates domain-specific knowledge to mitigate LLMs' limitations in biomedical categorization. Experiments across four BioNER datasets show that SKIPNER significantly outperforms prior LLM-based few-shot methods and matches the performance of other approaches that require up to fifty examples. However, the method incurs higher computational costs due to external knowledge retrieval, is sensitive to the quality of span extraction, and does not yet surpass fully supervised methods. By improving the accuracy and reliability of extracting biomedical entities, SKIPNER can directly support clinical decision-making by providing more precise information for differential diagnosis, treatment planning, and literature-based recommendations. To further advance BioNER via LLMs, future work on improving knowledge retrieval, refining span extraction, and leveraging instruction tuning for better entity classification was suggested by the researchers[71].

Patient-centered applications demonstrate LLMs' impact beyond clinical reasoning. ChatGPT-4 was rated more empathetic and useful than physician responses by patients, though some outputs carried hidden safety risks[72]. MindfulDiary guided patients with major depressive disorder in journaling, enhancing self-reflection and clinician insight[73]. Mental-LLM studies indicate instruction-fine-tuned LLMs outperform zero- and few-shot methods in mental health prediction but require domain-specific tuning and further validation to mitigate ethical risks[74]. TwIPS and similar tools support autistic individuals in social communication, improving clarity and reducing emotional risk, though expert oversight remains important[75]. Wheelpedia, a WhatsApp-based chatbot, demonstrates LLM potential for scalable support in resource-limited settings, highlighting the need for localization and co-design[76]. LLMs

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have also been applied to elderly patient Question & Answering (Q&A)[77] and glaucoma-related queries[78].

The persuasive potential of AI-generated content for public health communication was studied by a group of researchers from the University of Georgia and Stanford, focusing on messages promoting COVID-19 vaccination. Using GPT-3, the authors conducted a three-part investigation into the effectiveness of LLMs in generating impactful public health messages. In Study 1, the team systematically evaluated GPT-3's ability to generate coherent and relevant pro-vaccination messages. Study 2 compared these AI-generated messages to official Centers for Disease Control and Prevention (CDC) messages, revealing that participants perceived the GPT-3 messages as more persuasive, more strongly argued, and eliciting more positive attitudes than those from the CDC. Study 3 explored how message source labels affect perception: although participants preferred the AI-written content, they reacted negatively when messages were explicitly labeled as AI-generated. These findings highlight the dual nature of the role of AI in public health communication. LLMs like GPT-3 can enhance message effectiveness, but source credibility remains crucial for public trust[79].

LLMs can also assist in triage and referral decision-making by integrating patient-reported symptoms with medical knowledge to suggest urgency and appropriate specialist care. For instance, a recent study benchmarked multiple LLMs and an LLM-based workflow using retrieval augmented generation (RAG) on 2,000 cases from the Medical Information Mart for Intensive Care (MIMIC) database. The models successfully provided personalized insights into likely diagnoses, recommended specialists, and assessed urgent care needs, supporting clinicians in refining decisions and prioritizing care[80]. **Figure 9** illustrates an example use case of LLM-assisted triage: a patient presenting with chest pain and shortness of breath is evaluated using zero-shot, few-shot, and COT prompting strategies, ultimately classifying the case as Urgent. This demonstrates how LLMs can function as a real-time clinical decision support tool, complementing clinician expertise and improving patient care efficiency.

Drug discovery and development and DNA

LLMs are accelerating progress in drug discovery and development, as well as in the analysis of DNA, RNA, and protein sequences, by enabling rapid interpretation of scientific literature, prediction of molecular interactions, and generation of novel hypotheses. Their ability to treat biological sequences as a "language" has paved the way for innovative approaches in target identification, compound screening, and genomic analysis. In parallel, domain-specialized architectures, such as protein language models (PLMs) and genomic language models (GLMs), extend these capabilities to the molecular level, learning biochemical and structural rules directly from sequence data. PLMs, for instance, have been applied to protein structure prediction, functional annotation, and rational protein design, enabling more informed drug target

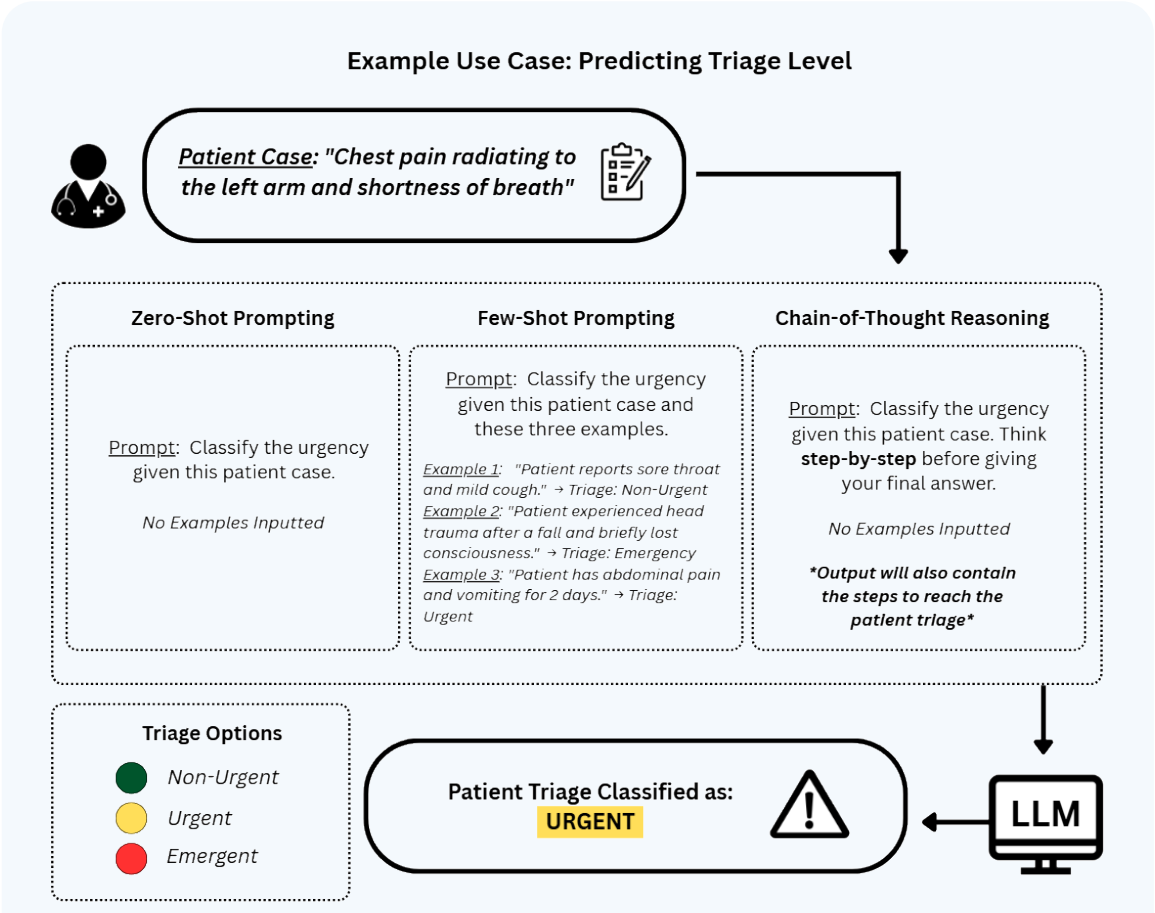


Figure 9 Example use case of LLMs for predicting patient triage level. A patient case describing chest pain radiating to the left arm and shortness of breath is evaluated using three prompting strategies: (1) Zero-shot prompting—classifying without prior examples; (2) Few-shot prompting—classifying with several annotated examples; and (3) Chain-of-thought reasoning—generating a step-by-step explanation before providing the final classification. Based on the LLM's reasoning, the case is classified as Urgent from among three triage options: Non-Urgent, Urgent, and Emergent.

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validation and the creation of novel therapeutic candidates. Similarly, GLMs facilitate variant effect prediction, regulatory element discovery, and the design of synthetic DNA or RNA sequences, directly supporting precision medicine pipelines.

ClinicalAgent is a multi-agent clinical trial assistant that integrates GPT-4 with advanced reasoning methods such as ReAct and LEAST-TO-MOST to improve clinical trial outcome prediction and management. Mimicking a hospital staffed by specialized agents (e.g., for pharmacology, diagnosis, or trial design), the system enables explainable, domain-specific reasoning by routing tasks to the appropriate GPT-4-powered agent. ClinicalAgent enhances LLM performance by combining predictive analytics, failure analysis, and trial duration estimation, achieving a Precision-Recall Area Under the Curve (PR-AUC) of 0.7908, outperforming standard prompt-based approaches by 0.3326. It also features a web interface aimed at fostering community collaboration. As the first multi-agent LLM system for clinical trials, ClinicalAgent establishes a novel paradigm in AI-powered medical research support. However, despite its strong performance, ClinicalAgent currently depends on human-led agent configuration, limiting scalability. Future work will focus on enabling self-adaptation and dynamic learning from interaction[81].

DrugAssist is an interactive molecule optimization model that leverages LLMs to perform molecular property optimization through human–machine dialogue, addressing the limitations of prior noninteractive approaches. By integrating expert feedback iteratively, DrugAssist improves the optimization of both single and multiple molecular properties, including challenging range-specific targets. Evaluated across 16 tasks, DrugAssist consistently outperformed popular LLMs like GPT-3.5 and LLaMA2 in validity and success rates of generated molecules. The model's effectiveness is further supported by the release of MolOpt-Instructions, a large instruction-based dataset for fine-tuning LLMs on molecule optimization[82]. Drug repurposing offers an efficient and cost-effective pathway in drug discovery, yet traditional models often struggle to generalize across the vast molecular and target spaces due to limited data and model capacity. The DrugReAlign framework uses LLMs combined with multi-source prompting, including target summaries and spatial interaction data, to enhance drug-target interaction (DTI) prediction and repurposing accuracy. Validated through molecular docking and DTI datasets, DrugReAlign demonstrates improved predictive performance and interpretability compared to conventional approaches. Importantly, the study reveals a strong correlation between the semantic quality of LLM-generated drug explanations and docking outcomes, underscoring the importance of precise prompt design. Despite challenges such as LLM hallucinations when data is sparse, integrating diverse, high-quality data sources mitigates these issues and improves robustness[83]. Molecular generation is a key AI-driven technology accelerating small molecule drug development, but existing language model approaches often face challenges in handling diverse drug design tasks like linker design due to their autoregressive nature. To address this, FragGPT, a universal fragment-based molecular generation model, was introduced, leveraging an unordered fragment representation called FU-SMILES. Pretrained on large molecular datasets, FragGPT efficiently supports a variety of design tasks, including de novo molecule generation, linker design, scaffold hopping, and side chain optimization. By incorporating conditional generation and reinforcement learning, FragGPT produces molecules with multiple optimized biological and physicochemical properties. Experimental validations demonstrate its superior ability to generate novel, property-enhanced molecules compared to current SOTA models, with successful application in real-world drug design scenarios[84].

Predicting blood–brain barrier (BBB) permeability of small molecules is critical in drug discovery, particularly for central nervous system therapies. Recent advances leverage AI, combining transformer-based language models and machine learning to enhance prediction accuracy and speed. One such approach uses the transformer-based MegaMolBART model to encode molecular SMILES representations, paired with an XGBoost classifier, outperforming traditional fingerprint-based methods. This hybrid in silico model achieved a high predictive performance (AUC = 0.88) on test datasets and was experimentally validated using 3D human BBB spheroid models. Results confirmed that predicted BBB-permeable compounds successfully crossed the barrier, while impermeable compounds did not, demonstrating strong concordance between computational and laboratory data[85].

Fine-tuned LLMs are emerging as powerful tools for mining complex chemical literature, a long-standing challenge in both experimental and computational chemistry. Zhang et al. evaluated the performance of LLMs across five critical chemical text mining tasks: compound entity recognition, reaction role labeling, metal-organic framework (MOF) synthesis information extraction, nuclear magnetic resonance (NMR) data extraction, and conversion of reaction paragraphs into structured action sequences. While models like GPT-3.5-turbo and GPT-4 were evaluated via advanced prompt engineering, the study primarily focused on fine-tuning open-source models such as Mistral, Llama3, T5, and BART. These fine-tuned models achieved exact-match accuracies ranging from 69% to 95%, in many cases outperforming task-adaptive models trained on large in-domain datasets. These results highlight the efficiency and adaptability of fine-tuned LLMs for chemical information extraction, particularly when annotated data is scarce[86].

General healthcare and surgery

LLMs are emerging as intelligent assistants in healthcare and surgery, supporting record interpretation, documentation, and team communication. Surgical applications include preoperative planning, intraoperative support, and postoperative monitoring. In frontline care, they are being tested for triage, symptom checking, and operational optimization. This section surveys their integration into clinical and surgical contexts, highlighting contributions to safety, efficiency, and innovation.

Trujillo et al. in Chile evaluated GPT-4 against experienced surgeons in providing feedback on surgical scenarios. Neutral case narratives were reviewed by ChatGPT and three surgeons, with feedback judged by residents and experts. ChatGPT's responses were rated useful in over 96% of cases, comparable to surgeons, and often indistinguishable from human feedback. It also identified errors at rates similar to surgeons[87].

Mahmoud et al. assessed four LLMs (GPT-4o, GPT-3.5, Gemini, and Copilot) on 714 oral and maxillofacial surgery (OMS) board exam questions. GPT-4o achieved the highest accuracy (83.7%) and corrected 98.2% of initial errors, outperforming others across all 11 OMS domains. Results indicate strong promise for OMS education, though domain variability requires refinement before widespread use[88].

Stengal et al. tested ChatGPT-3.5, Bing, and Bard on 86 questions from the European Board Exam in Neurological Surgery. Bard performed best, answering 62% correctly overall and 69% excluding image-based items. It surpassed humans on theory-based content (86% vs. 60%, $p = 0.03$), though all LLMs failed entirely on image-based questions[89].

To improve patient comprehension, NYU Langone researchers used ChatGPT-4 to simplify MRI reports for orthopedic surgery patients. In a randomized trial, patients receiving AI-translated knee MRI reports showed a 20% comprehension gain, and 87.5% preferred the AI version for clarity, though trust was slightly lower compared to radiologist reports. Findings highlight LLMs' potential to enhance understanding of complex imaging with necessary clinical oversight[90]. Another study compared ChatGPT and Bard in patient education on lumbar disc herniation. Spine surgeons rated over 70% of responses as excellent or satisfactory, citing clarity and empathy, though occasional inaccuracies and rater variability underscored the need for oversight and regulatory safeguards[91].

ChatGPT's role in otolaryngology-head and neck surgery was assessed using the Concordance, Validity, Safety, Competency (CVSC) framework. It achieved a passing score (75%) and improved with guided prompts. While results showed high concordance and validity, concerns about partial accuracy and hallucinations emphasized the need for refinement

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before deployment[92]. Soroudi et al. explored LLMs and text-to-image tools for education in hand surgery. ChatGPT-4.5 outperformed ChatGPT-3.5 on readability and patient-centered metrics (accuracy 4.47/5; understandability 91.7%; actionability 54.7%). In parallel, GPT-4o generated the most accurate images compared to DALL-E 2 and DALL-E 3, underscoring its multimodal value[93].

LLMs are also being tested for automating administrative tasks like current procedural terminology (CPT) coding. In a study of hand surgery cases, five LLMs were compared. Perplexity.AI performed best on simple and complex procedures, while Bard and Bing AI produced mixed results. ChatGPT models underperformed, particularly on complex tasks, though Bing AI generated the most partially correct responses[94].

Healthcare education

LLMs are playing an instrumental role in healthcare education by personalizing learning pathways, simulating patient interactions, and providing real-time answers to complex medical queries. Educators are increasingly using LLMs to enhance curricula and support continuous professional development. This section highlights LLM capabilities across standardized and nonstandardized assessments, patient education, and specialty-specific knowledge, illustrating their potential as educational tools while noting current limitations.

All physicians with a Doctor of Medicine (MD) degree in the United States are required to take the Medical Licensing Exam (USMLE) sponsored by the Federation of State Medical Boards (FSMB) and National Board of Medical Examiners (NBME)[95]. In 2022, ChatGPT's performance on Steps 1 and 2 of the exam was evaluated, and based on its performance on the NBME FreeStep 1 dataset, researchers concluded the LLM performed at the level of a third-year medical student. Perhaps more revealing of its ability to assist in healthcare education, ChatGPT was able to provide reasoning and context for the majority of its answers[96].

ChatGPT has also been evaluated on professional medical exams outside of the English language. In a study conducted evaluating GPT 3.5 on the Japan National Nurse Examinations on all exam questions from 2019 to 2023, ChatGPT answered 75.1% of basic knowledge questions (questions that test basic issues of particular importance in the nursing field) and 64.5% of general questions (questions that test a wider variety of and more specialized knowledge compared to basic questions) on average over the 5 years. Overall, using GPT 3.5, the model based on the 2019 exam came within a few questions of passing the 2020–2023 exams. Additionally, the model performed better in certain areas, such as nutrition and pathology, than in areas such as social welfare and law[97]. Also in Japan, GPT-4 outperformed average medical residents on the General Medicine In-Training Examination (GM-ITE) in Japanese. The mean score for GPT-4, 70.1%, was significantly higher than the mean score of residents tested, 55.8%, using a p-value of 0.001. However, analogous to findings in research studies that explored LLMs as potential tools for research drafting, ChatGPT performed lower than medical residents in areas involving professionalism[98], while scoring higher in areas such as specific diseases[99].

The accuracy of ChatGPT versus Bing Chat was also explored in answering questions on the Japanese National Medical Licensure Examination (NMLE). A total of 295 candidate questions formed the initial pool, with 14 being excluded given that Bing Chat referred to a website where a correct answer was already posted, leaving 281 questions. Of the 281 questions used to compare the two models, 81 were compulsory questions and 200 were categorized as other questions. The researchers found that Bing Chat outperformed ChatGPT in terms of passing the exam with an overall score of 78% versus 38%, and most of Bing Chat's incorrect answers stemmed from wrong information, information that could not be determined as incorrect without prior knowledge. Although it outperformed ChatGPT in this study, Bing Chat was noted to still have limitations with a 20% error proportion, and the researchers stressed the need for professionals to check the references for the Bing Chat searches before assuming it is true to mitigate the error proportion rate[100].

Taking the Chinese National Medical Licensing Examination (NMLE), researchers used GPT-4 and found that the model achieved a score of 442/600, above the required passing score of 360/600, for questions that were kept in Chinese. When they tested the model against questions translated into English, the model achieved similar results. However, the model showed limitations in open-ended questions, where it only achieved 47.7% accuracy. Despite the limitations in answering the open-encoded exam questions, the researchers concluded that ChatGPT's ability to perform comprehensive reason is apparent on the Chinese NMLE, further bolstering the profile of LLMs as education resources while noting that the utility of such models should be studied further in real-world conditions given the complexity of clinical tasks that can't be fully simulated in an exam sitting[101].

In Spain, the Medical Intern (MIR) exam is required to train in the National Health System toward becoming a specialist after medical school. In total, there are 47 registered specialties, each requiring 4 to 5 years as an intern to pursue. The exam is made up of 200 multiple-choice questions and 10 reserve questions to address question formulation issues, with the overall exam grade carrying a weight of 90% toward specialty placement. The 2022 exam was administered in a study conducted by Guillen-Grima et al., using 182 of the exam questions made up of two categories, theoretical and practical, after removing 25 questions that were related to an unavailable image and 3 challenge questions, to both GPT-4 and GPT-3.5. In Spanish, GPT-4 answered 86.81% of questions correctly compared to 63.18% for GPT-3.5. The English translation of the questions was also evaluated, with GPT-4 scoring 87.91% and GPT-3.5 66.46%. GPT-4 performance on the Spanish question set was similar in both theoretical and practical categories, scoring 87.1% and 86.6%, respectively, in 19 of the 35 specialities, consisting of 81 total questions, GPT-4 answered related questions 100% correctly. Overall, the authors did not suggest the models were ready for reliable deployment in a healthcare setting without proper oversight, but the potential and robust capabilities of the LLM were evident[102].

ChatGPT has also shown the ability to accurately answer questions for questionnaires associated with particular diseases and fields. The 24-DKQ, a 24-item diabetes knowledge questionnaire, was presented to ChatGPT, and accurate responses were provided for each question, showing the potential to transform diabetes education and self-management[103]. In the field of ophthalmology, researchers used two 260-question multiple-choice datasets from the Basic and Clinical Science Course (BCSC) Self-Assessment Program and OphthoQuestions, comparing the performance of GPT-4 and GPT-3.5 with each other and to humans. The top-performing model, GPT-4 with a temperature setting of 0.3, performed best with a 75.8% accuracy on the BSCS questions and 70.0% accuracy on the OphthoQuestions questions, 2.2% and 7.0% higher than humans on each question set, respectively, although the differences were not statistically significant[104]. Mondal et al. evaluated ChatGPT's effectiveness in providing answers on lifestyle-related diseases such as diabetes and obesity. Across 20 cases with four questions each, ChatGPT demonstrated reasonably accurate information with an average accuracy score of 1.83 and a guidance score of 1.9, surpassing the hypothetical median of 1.5, as judged by two physicians. The findings suggest ChatGPT can offer initial guidance to patients on health queries before consulting a doctor[105]. In the field of orthopedics, researchers in Australia found that implementing ChatGPT-4 into orthopedic education could reduce experience gaps and other knowledge shortcomings that junior clinicians suffered from as a result of the COVID-19 pandemic[106].

Outside of standardized exams, LLM performance has also been evaluated in medical contexts. Researchers in Tunisia compared ChatGPT-3.5's performance on a French-language pulmonology exam to that of 244 third-year medical students. In its basic form (V1), ChatGPT showed strong performance in factual areas like radiology and

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microbiology but struggled in applied clinical domains. When provided with contextual cues (V2), ChatGPT's accuracy improved significantly across most topics and question types. V2 surpassed 62.1% of students overall, particularly excelling in low-to-moderate difficulty questions and open-ended formats. However, ChatGPT underperformed on multiple-choice items and complex cases requiring clinical reasoning. These results suggest LLMs can complement medical education by reinforcing knowledge recall and interpretation but still lack deeper clinical judgment and reasoning, especially when tested in non-English settings or under high-cognitive-load scenarios[107].

Educating patients is also an area of interest explored using LLMs. Scquizzato et al., in tandem with Sudden Cardiac Arrest UK, produced a list of 40 questions related to cardiac arrest and cardiopulmonary resuscitation (CPR) to assess whether ChatGPT could answer questions about the topics from 14 professionals and 16 nonprofessionals. The results showed that the model was evaluated positively overall using a 5-point scale, receiving an average rating of 4.3, including 4.4 on clarity, 4.3 on relevance, 4.2 on comprehensiveness, and 4.0 on accuracy. Notably, professionals rated overall value lower (4.0 vs. 4.6) as well as comprehensiveness (3.9 vs. 4.5), and CPR-related question responses received lower evaluations from both groups across all metrics. Ultimately, the study led the researchers to conclude that ChatGPT was able to provide satisfactory answers across all metrics for cardiac arrest except the CPR-related questions and that LLMs will play a significant role in healthcare going forward[108].

Ethics, a key component of healthcare education, has also been studied using ChatGPT. Chen et al. performed a study assessing GPT 3.5's accuracy in answering bioethics questions at the level of third-year and fourth-year U.S. medical students, revealing an overall accuracy of 59.6%. While it performed well in certain topics and specialties such as pediatrics, notable shortcomings were observed, emphasizing the need for further research to comprehend the capabilities and limitations of AI systems in medical bioethics[109]. Table 4 provides a summary of select medical examinations taken by LLMs.

Table 4 Select medical examinations taken by LLMs.

LLM	Exam	Score	Passed	Language	Date
ChatGPT	Chinese NMLE	442/600	Yes	Chinese	Dec 2023
ChatGPT (GPT-4)	BCSC	75.8%	Yes	English	Nov 2023
ChatGPT (GPT-4)	Ophtho Questions	70.0%	Yes	English	Nov 2023
ChatGPT (GPT-4.0)	Spanish MIR	86.61%	N/A	Spanish	Sep 2023
ChatGPT (GPT-4.0)	Spanish MIR	87.91%	N/A	English	Sep 2023
ChatGPT (GPT-3.5)	Japanese NNE (2023)	71.7%	No	Japanese	Jun 2023
ChatGPT (GPT-3.5)	Japanese NNE (2022)	77.1%	No	Japanese	Jun 2023
ChatGPT (GPT-3.5)	Japanese NNE (2021)	73.9%	No	Japanese	Jun 2023
ChatGPT (GPT-3.5)	Japanese NNE (2020)	72.9%	No	Japanese	Jun 2023
ChatGPT (GPT-3.5)	Japanese NNE (2019)	80.0%	Yes	Japanese	Jun 2023
ChatGPT	AMBOSS-Step1	44% (44/100)	No	English	Feb 2023
ChatGPT	AMBOSS-Step2	42% (42/100)	No	English	Feb 2023
ChatGPT	NBME-Free-Step1	64.4% (56/87)	Yes	English	Feb 2023
ChatGPT	NBME-Free-Step2	57.8% (59/102)	No	English	Feb 2023
Bing Chat	Japanese NMLE	78.0%	Yes	Japanese	Feb 2022
ChatGPT	Japanese NMLE	38.0%	No	Japanese	Feb 2022

Scientific literature analysis

The rapidly expanding body of healthcare and biomedical informatics literature poses challenges for clinicians and researchers striving to remain current. LLMs help mitigate this information overload by summarizing findings, identifying trends, and enabling intelligent search and question answering. They increasingly act as literature navigators and research copilots, uncovering insights from vast publications.

A growing area of interest is research paper drafting. Macdonald et al. generated a synthetic dataset of 100,000 healthcare workers with varied phenotypes and risk profiles, a subset receiving a simulated vaccine that reduced hospitalization risk. After describing the dataset to ChatGPT, the model suggested experimental designs, provided R code for statistical analysis, and drafted a manuscript with standard sections. The authors concluded that ChatGPT can assist with study design, analysis, and drafting, though it cannot replace human oversight[110]. Wen and Wang similarly noted ChatGPT's usefulness for proofreading and manuscript checking but highlighted inaccuracies and ethical concerns that limit uncritical adoption[111]. Other proposed applications include automating visual elements such as figures and tables, which are essential but time-consuming to prepare[112].

The LLMs4Synthesis framework was designed to improve LLM summarization of complex literature. It introduces novel synthesis types, nine quality criteria, and reinforcement learning with AI feedback (RLAIF) to align outputs with these metrics. A notable innovation is using LLMs both as generators and evaluators of summary quality, with GPT-4 assessments surpassing traditional metrics like BERTScore and BLEURT. The framework also integrates a rule-based paper structure identifier to improve format adherence. Experiments showed that RLAIF optimization produced more reliable and higher-quality summaries than baselines. By releasing resources publicly, the authors support continued application of LLMs to bioinformatics and related domains[113].

LLMs have also been tested for generating lay summaries. For example, ChatGPT-3.5 created patient-facing versions of vascular surgery abstracts, achieving higher readability scores and lower grade-level requirements than the originals. However, expert review found factual inaccuracies, with one surgeon identifying errors in nearly one-third of summaries and poor interrater agreement ($k = -0.1$). These findings suggest LLMs can improve accessibility but require expert validation to ensure accuracy[114].

Finally, LLMs have been evaluated for literature scoring. For example, Roberts et al. tested ChatGPT-3 on implant dentistry abstracts using the CONSORT-A checklist, comparing overall compliance scores with human evaluation. Bland–Altman analysis showed a mean difference of 4.92%. While token limitations affected ChatGPT-3, the authors noted likely improvements with ChatGPT-4 and suggested integrating LLMs with medical databases to enhance accuracy[115].

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EVALUATION BENCHMARKS AND METRICS

Rigorous evaluation of LLMs in healthcare is critical for assessing their reliability, safety, and clinical utility. Unlike traditional NLP tasks, clinical applications demand not only high accuracy but also trustworthiness, interpretability, and alignment with medical standards. Several benchmarks and metrics have emerged to evaluate LLM performance across different healthcare domains.

Clinical question answering and reasoning

Benchmarks such as MedQA (USMLE-style medical questions)[116] and PubMedQA[117] test LLMs on their ability to reason over clinical knowledge and medical literature. Metrics typically include accuracy, F1-score, and exact match, providing a straightforward measure of correctness. These benchmarks help identify LLM strengths in knowledge retrieval and reasoning but are limited by their focus on factual recall rather than patient-specific recommendations.

Table 5 Overview of select biomedical benchmarks.

Benchmark Name	Task Type	Domain	Limitations
BLURB[29]	Multi-task: NER, relation extraction, QA, classification, PICO, similarity	Biomedical literature-based NLP (PubMed, clinical, scientific text)	Text-based only; covers many tasks but not molecular generation; some tasks have limited data
BioASQ[122]	Biomedical QA (factoid, list, summary), semantic indexing	Biomedical research, literature review	Focused on QA from literature; requires external knowledge sources; not molecular or code tasks
BioCoder[123]	Code generation (Python, Java) for bioinformatics tasks	Bioinformatics, computational biology	Focused on code, not natural language; domain-specific coding knowledge required
BioinformaticsBench[124]	QA, problem-solving, biomedical reasoning	Genetics, genomics, proteomics, metagenomics, single-cell analysis	Focused on collegiate-level problems; may not cover all real-world scenarios; error analysis shows varied LLM strengths
BenchHealth[125]	Clinical reasoning, generation, comprehension	Clinical applications	Heterogeneous tasks; may lack depth; gap between LLM and clinical needs
DNALongBench[126]	Long-range DNA sequence prediction for genomics tasks	Genomics, regulatory genomics	Focused on modeling long-range dependencies; limited to DNA sequence tasks; not for small molecules or clinical text
GuacaMol[127]	Molecular generation, property optimization, distribution learning	Molecular design, drug discovery	Focuses on small molecule generation; limited biological context; may reward trivial diversity; not designed for LLMs specifically
MedExQA[128]	QA for five medical specialties	Biomedical engineering, lab sciences	Limited number of specialties
MedMCQA[129]	Multiple-choice QA from Indian medical exams	Clinical knowledge, medical education	Regional focus; MCQ only; deep reasoning but exam context
MedQA[116]	Multiple-choice QA for USMLE questions	Clinical decision-making, medical education	Lacks real patient context; focused on USMLE; mostly English
MedQA-CS[130]	OSCE QA	Clinical reasoning	Newer benchmark with limited adoption
MedS-Bench[131]	Multi-task: QA, summarization, NER, etc.	Clinical applications	Expensive to evaluate at scale; broad but shallow per task
MedXpertQA[132]	QA for medical image reasoning and advanced clinical case reasoning	Clinical reasoning	Higher computational demands due to multimodal data types
MIMIC-III/IV[118]	Prediction, extraction, and classification from EHRs	Critical care, hospital data	Noisy, complex real-world data; needs domain expertise
MOSES[133]	Molecular generation, scaffold-based evaluation, property prediction	Molecular design, drug discovery	Focused on chemical structure generation; limited to small molecules; not tailored for biological tasks; may not test reasoning behind generated chemicals
MoleculeNet[134]	Molecular property prediction	Cheminformatics, drug discovery	Graph-based; predates LLMs; limited to molecular properties
MMLU[135]	General multitask multiple-choice QA	General knowledge (includes medical)	Broad but shallow; not specialized for medicine
MultiMedQA[136]	Multi-format QA (MCQ, long-form)	General healthcare, medical education	Reference answers are not always clinician-verified
PubMedQA[117]	Factoid QA based on scientific abstracts	Biomedical research, literature comprehension	Limited to abstracts; simple answer types; minimal clinical context
TRINDs[137]	Real-world epidemiological data aggregation	Epidemiology, global health	Licensed access; heterogeneous data; not for model evaluation

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Electronic Health Record tasks

Datasets like MIMIC-III/IV[118] evaluate tasks including clinical note summarization, NER, and temporal reasoning over patient records. Metrics often include AUROC, precision, recall, and F1-score. These evaluations assess not only predictive performance but also the model's ability to handle noisy, sparse, or longitudinal clinical data.

Multimodal and imaging benchmarks

For tasks that combine text and imaging, such as radiology report generation, benchmarks include MIMIC-CXR[119] or custom radiology QA datasets. Metrics such as Bilingual Evaluation Understudy (BLEU)[120], Recall-Oriented Understudy for Gisting Evaluation (ROUGE)[121], and Consensus-based Image Description Evaluation (CIDEr) assess the quality of generated text, while domain-specific evaluations may involve clinical correctness scoring by expert radiologists. Multimodal evaluation remains a challenging area due to the difficulty of aligning model outputs with both textual and visual clinical standards.

Safety, hallucination, and calibration metrics

Given the high stakes of clinical decision-making, evaluation has expanded beyond traditional accuracy to include hallucination detection, uncertainty calibration, and fairness metrics. Techniques such as self-consistency checks, expert validation, and model confidence scoring help quantify the trustworthiness of LLM outputs, which is crucial for safe deployment in healthcare.

Limitations and future directions

Current benchmarks are often institution- or dataset-specific, limiting generalizability. There is a growing need for standardized, multi-institutional, multimodal benchmarks that capture real-world clinical variability. Additionally, metrics that integrate clinical relevance, patient safety, and ethical considerations will become increasingly important as LLMs move from research to deployment. An overview of select datasets and benchmarks is shown in Table 5.

CHALLENGES

While LLMs present transformative opportunities in healthcare and biomedical informatics, their safe and responsible deployment faces major obstacles. These challenges span privacy and security, data quality, explainability, generalizability, regulatory and ethical issues, workflow integration, computational costs, and continuous adaptability. Taken together, the challenges we discuss convey three key messages: (1) without careful attention to bias, privacy, and data governance, LLM deployment risks amplifying existing inequities in healthcare; (2) achieving explainability and clinician trust is as critical as model performance; and (3) ethical, legal, and infrastructural barriers remain central obstacles that must be overcome before LLMs can move from promising pilots and research studies to safe, scalable integration in routine clinical practice. Figure 10 highlights the key challenges of LLMs in healthcare and biomedical informatics discussed in this section.

Data privacy and security

Healthcare data is highly sensitive and regulated under frameworks such as Health Insurance Portability and Accountability Act (HIPAA) and General Data Protection Regulation (GDPR). Fine-tuning on patient records risks model memorization or inadvertent disclosure of protected health information (PHI), even with de-identification. Multimodal and longitudinal datasets heighten reidentification risks. Approaches such as federated learning, differential privacy, and synthetic data generation are being explored, but scalable solutions remain limited. RAG offers a promising alternative by enabling secure inference over external knowledge bases without retraining. However, RAG itself requires strong safeguards for data storage, access control, and filtering, since retrieval systems may still surface sensitive content[138–144].

Lack of high-quality, domain-specific datasets

LLMs require large, diverse corpora, yet biomedical datasets are scarce, fragmented, and often biased or noisy. Ethical constraints further restrict availability, limiting adaptation to tasks such as phenotype extraction, variant interpretation, and EHR summarization. Synthetic data provides one workaround, enabling training without compromising privacy. For example, an LLM-based tool for detecting thrombolysis contraindications used 150 synthetic stroke notes, achieving >90% sensitivity and nearly 99% specificity with no hallucinations[145]. While promising, synthetic data remains complementary and requires validation against real-world datasets before clinical deployment.

Explainability and interpretability

For LLMs to gain clinical acceptance and eventual implementation, their generated outputs must be explainable to clinicians and researchers. Unlike rule-based systems, LLMs operate as opaque black boxes. The lack of transparency introduces barriers in high-stakes environments such as diagnosis and treatment planning, where decisions require clear justification. Efforts to improve model interpretability through mechanisms such as attention visualization, saliency maps, and counterfactual explanations are ongoing but have not reached the scope and reliability needed for widespread acceptance in the clinical community. Even as hindrances to explainability and interpretability are reduced, adoption and acceptance will still likely face barriers based on clinician experience. For example, a recent nationwide study conducted in Greece explored how medical professionals interact with ChatGPT-4.0 in real-world clinical settings. The study involved 182 physicians from various specialties and experience levels, who used a custom web application interfacing with ChatGPT over a one-month period. Doctors rated their experiences across four key dimensions: clarity of responses, response time, informational accuracy, and overall satisfaction. ChatGPT's response time was highly rated (3.85/5), while clarity and accuracy received more moderate evaluations (around 3.43/5). However, the study identified a negative correlation between years of experience and satisfaction levels: physicians with over 21 years of experience reported the lowest satisfaction (2.80–3.74), whereas younger doctors rated the system more favorably (3.43–4.20). Explainability and interpretability will be critical to further LLM acceptance in the clinical domain and will likely have an even higher threshold for more experienced practitioners with decades of knowledge and training[146].

Generalizability and bias

LLMs trained on general or region-specific corpora often fail to generalize across institutions, populations, or clinical contexts, risking inequities in care. For instance, GPT-4 showed strong sensitivity in detecting suicidal ideation but weaker precision and poorer performance when predicting future crises, reflecting limited generalizability. Evaluation itself can introduce bias: human judgments are subjective[147,148], and automated metrics may inherit dataset biases. Standardized, transparent evaluation frameworks combining human expertise with robust computational metrics are essential for equitable deployment.

Regulatory and ethical considerations

Biomedical data often combine genomic, phenotypic, and environmental information[149], raising challenges around consent, secondary use, and provenance. Existing governance frameworks designed for narrow studies struggle to accommodate LLM scale and flexibility. Model opacity further complicates auditability in areas such as genomic diagnostics or treatment prioritization. Ethical deployment requires frameworks ensuring traceability, transparency, and ongoing human oversight.

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Figure 10 Key challenges in deploying LLMs in healthcare and biomedical informatics. Challenges span data privacy and security, dataset quality, explainability, generalizability, ethics, workflow integration, computational cost, and continual adaptability. Addressing these areas is critical for safe, effective, and responsible adoption.

Integration with clinical workflows

Clinical adoption depends on seamless workflow integration, which is hindered by heterogeneous EHR systems[150], time constraints, and cognitive load. LLM outputs must be contextually relevant, real-time, and easy to use across technical backgrounds[151]. Poorly designed systems risk adding alerts and manual corrections, exacerbating burnout[152–157]. A notable success is an NLP-based EHR message routing system, which reduced staff workload and improved response times while maintaining high accuracy[158]. Such cases show LLMs can streamline workflows, but broader success requires user-centered design and interoperability.

Computational cost and environmental impact

Training and inference of LLMs requires substantial energy. For instance, GPT-3's training alone reportedly emitted 552 metric tons of CO_2, roughly equivalent to the emissions from 300 transcontinental flights[159]. Beyond training, inference at scale can also be energy-intensive: some models consume over 70 times the energy of GPT-4.1 Nano, with annual usage projected to equal the electricity consumption of 35,000 U.S. homes and the water usage of 1.2 million people[160]. To mitigate these environmental impacts, a combination of algorithmic and infrastructure strategies can be employed. Algorithmic optimizations include quantization[161], pruning[162], distillation[163], and mixture-of-experts architectures[164]. Infrastructure improvements involve the use of energy-efficient AI hardware[165], advanced cooling systems[166], and sourcing renewable energy[167].

Continuous learning and adaptability

Biomedical knowledge evolves rapidly, requiring LLMs to update continuously while avoiding catastrophic forgetting[168]. Frequent retraining is costly and impractical, while fine-tuning risks inconsistency. Incremental and modular strategies, combined with robust validation pipelines and human-in-the-loop curation, may balance stability with adaptability. Adaptability must also extend to shifting clinical workflows and evolving user needs, suggesting hybrid approaches that integrate validated static knowledge with dynamically updated modules.

FUTURE DIRECTIONS

The future of LLMs in healthcare and biomedical informatics will hinge on overcoming current limitations and broadening clinical utility. Three key messages emerge from the analysis of future LLM directions in the space: (1) the next generation of LLMs will depend on multimodality, memory, and multi-agent coordination to handle the full complexity of clinical data and workflows; (2) their successful translation into practice requires careful integration with immersive technologies, robotics, and real-time systems, supported by interdisciplinary collaboration among computer scientists, clinicians, and policymakers; and (3) responsible deployment must embed compliance, explainability, and equity by design, ensuring that technical advances are matched with ethical and regulatory safeguards.

Integration with immersion technologies

Combining multimodal LLMs with immersive technologies offers new opportunities in healthcare and biomedical informatics[169]. For aging populations, systems like ElderEase AR enable elderly users to capture images, ask context-aware questions, and receive personalized feedback. A pilot study with 30 adults showed high usability and potential for independence and quality of life[170]. In clinical research, TWIN-GPT generates personalized digital twins from sparse EHRs to improve virtual trial outcome prediction. By preserving individual characteristics and synthesizing realistic data, it enables higher-fidelity simulations compared to prior models, especially

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in low-data contexts[171]. Together, these examples illustrate how immersive, multimodal LLM platforms can bridge the gap between digital intelligence and real-world clinical or daily-life applications, laying the foundation for more personalized, accessible, and effective healthcare interventions.

Multi-agent systems

Unlike single-model architectures, multi-agent systems coordinate multiple specialized AI agents, each with unique domain knowledge or functional capabilities, to simulate interdisciplinary collaboration, support complex reasoning, and deliver more context-aware clinical decision-making. These systems leverage the complementary strengths of LLMs, knowledge graphs, RAG, and domain-specific models to address limitations in scalability, accuracy, and transparency faced by traditional AI tools. By dynamically allocating tasks across agents and synthesizing diverse sources of information, multi-agent frameworks have the potential to transform diagnostic workflows, patient support tools, and medical research platforms.

In one example, researchers at Renmin University of China introduced a multi-agent medical decision support system that integrates a fine-tuned medical language model, biomedical knowledge graphs, and PubMed data retrieval to simulate interdisciplinary clinical collaboration. Designed to address the limitations of traditional healthcare QA systems, such as poor domain adaptability and limited inter-agent coordination, the system dynamically convenes specialized agents based on user-described symptoms to retrieve, synthesize, and deliver clinical recommendations. By combining multi-agent interaction with RAG, the system enhances diagnostic accuracy, credibility, and transparency. It was evaluated using both technical performance metrics and user-centric assessments, showing high user trust and satisfaction. Notably, clinicians responded positively to the system when it was framed as a supplementary source of evidence rather than a replacement for human judgment[172].

LLM advances

Enhancements in memory, fine-tuning, and knowledge integration will further extend clinical value. CareCall, a voice-based chatbot with long-term memory, improved user disclosure and perceived empathy, though selective memory design is required to balance privacy and comfort[173]. Parameter-efficient methods like MOELoRA, which combine Mixture-of-Experts with LoRA, enable scalable multi-task fine-tuning without full parameter updates, outperforming prior approaches on medical datasets[174]. Similarly, knowledge-tuning explicitly incorporates structured medical knowledge to reduce hallucinations and improve accuracy, as shown in the cMedKnowQA benchmark[175].

Other directions

LLMs also show promise in robotics, misinformation detection, and real-time clinical systems. In elder care, an LLM-powered social robot improved semantic fluency and cognitive engagement among retirement home residents, with strong acceptance for continued use[176]. TriIntel integrates deep neural networks, LLMs, and human input to detect complex health misinformation, outperforming stand-alone AI or hybrid baselines[177]. Systems like DrHouse demonstrate how real-time integration of wearable and sensor data with LLM consultations can improve diagnostic accuracy by 31.5%, while staying aligned with evolving medical knowledge[178].

Finally, regulatory compliance will be central. LLM deployment requires HIPAA/GDPR adherence, transparent explainability, and embedded audit and monitoring frameworks. Collaboration among AI researchers, clinicians, and legal experts will be key to building safe, trustworthy systems.

CONCLUSION

LLMs are transforming healthcare and biomedical informatics by enabling advanced understanding, generation, and reasoning over complex biomedical and clinical data. This review highlights how LLMs enhance scalability and automation in labor-intensive tasks, opening opportunities for personalized medicine and operational efficiency.

Broad adoption, however, faces challenges. Privacy and security require safeguards like federated learning, differential privacy, and RAG. Limited high-quality biomedical datasets constrain training, though synthetic data may help. LLM opacity raises explainability concerns, essential for clinician trust and regulatory compliance. Bias and poor generalizability underscore the need for rigorous evaluation with human oversight. Regulatory and ethical frameworks must ensure transparency, auditability, and patient consent. Integration is hindered by heterogeneous EHRs, clinician time constraints, and workflow disruption. High computational costs and environmental impacts demand algorithmic and infrastructure innovations, while continual learning is needed to keep pace with evolving biomedical knowledge.

Looking forward, multimodal LLMs integrating text, images, genomics, and sensor data can better capture clinical complexity. Immersive technologies and multi-agent systems may support aging populations, assistive care, and interdisciplinary collaboration. Future deployments must embed compliance, privacy, and explainability by design, requiring cooperation among AI researchers, clinicians, and policymakers. Addressing current limitations while advancing multimodality, continual learning, and responsible design, LLMs can augment expertise, accelerate discovery, and improve patient outcomes, driving a powerful, ethical, and equitable era of healthcare innovation.

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Human Rights Statements and Informed Consent: This article is a review of previously published studies and does not involve any new studies with human participants or animals performed by the authors. All procedures described in the included literature were conducted in accordance with the ethical standards of the respective institutional and/or national research committees and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards, as reported in the original publications. Informed consent was obtained by the original study investigators where applicable. No additional ethical approval or informed consent was required for this review.

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